

Actual theta sources and arithmetic metric extensions

Complete mathematical addendum

This addendum gives the complete current R67–R74 text and its final independent crosswalk; the full comparison of inherited endpoint-window transport with the later crossed-window refinement; the exact original-metric compactness calculation; and the full theta-admission and arithmetic metric-extension proofs, together with their independent reviews.

The selected sources appear in full. Their original coordinates, constants, masses, multiplicities, Taylor units, source maps and analytic domains remain attached to the formulas. Source text, editable LaTeX and exact conversion records accompany this reader. The earlier cumulative edition and its complete proof corpus remain separate preserved sources; this is an additive reader for the explicitly listed cohort.

Contents

Current results R67–R74	1
Complete independent crosswalk of R67–R74	5
Independent mathematical crosswalk: current Results R67–R74	5
R67 — actual τ pushforward and original arithmetic injections	5
R68 — coefficient family and rectangular period inclusion	5
R69 — arithmetic dilation, moving transport and reflection	5
R70 — full Taylor-unit formal gauge and contraction	6
R71 — analytic pole sequence, extension transport and exact endpoint	6
R72 — signed determinant derivative and curvature equality	7
R73 — all eigenvalues, source isometry and strictness scope	7
R74 — all-degree relation angles and nonlinear estimate	7
Scope of acceptance	9
Source pins and locators	9
Exact inherited EW, AGT and AA comparison	11
Exact comparison of inherited EW transport with AGT and AA	11
1. Exact source pins and reading scope	11
2. Original source, parameters, metrics and maps	12
3. Existing angle proof and the exact changed pairing	13
4. Full general-window consequence of the crossed pairing	15
5. Strict positivity: prior theorem and alternate later proof	16
6. The theta primitive is the same original cochain	17
7. Attribution text and cumulative integration scope	18
Projection compactness in the original metric	19
Theta-admitted analytic poles and original sources	20
Which analytic pole classes belong to the original theta source?	20
1. Original source, fixed parameters, and the exact norm	20
2. Complete Hilbert admission, including zeros on the observation line	20
3. Its entire cohomology and exact inclusion into the analytic family	21
4. Strong-Schwartz admission is an actual full-zero packet construction	21
5. Polynomial coordinates, source maps and every retained boundary defect	22
6. The actual theta Gram and finite source cost	23
7. The connection in the same arithmetic numerator frame	24
8. Exact recovery of the old canonical quotient metric	25
Sources and verification scope	26

Complete independent theta-admission proof	27
Added arithmetic directions and exact mixed volume	32
The added arithmetic directions: exact mixed metric and volume	32
1. Actual packets, original sources, and fixed coordinate conventions	32
2. Full block matrix and the exact mixed term	32
3. The original source quotient behind the Schur complement	33
4. Resultant, determinant line, and the full-unit conversion	33
5. The relation-volume is the original theta seed	34
6. Exact mixed-volume cost	34
Sources and verification scope	35
Complete independent metric-extension proof	36

Current results R67–R74

Result R67: the original pushforward over the absolute tau base. The complete original Tau Base construction works over the pointed blueprint $\mathbf{F}_{1,\tau} = \{\tau, 1\}$ and its four-point chart $+, - < \eta < \sigma$. In its declared coefficient-diagram category the projective resolution of the constant object is

$$0 \longrightarrow P_\eta \xrightarrow{v \mapsto (v, -v)} P_+ \oplus P_- \xrightarrow{(u,v) \mapsto u+v} \mathbb{C} \longrightarrow 0.$$

At the two lower points the last map is the identity; at the two upper points its kernel is the displayed anti-diagonal. This proves exactness pointwise. Applying Hom gives the actual derived pushforward

$$A_\tau(F) = [F_+ \oplus F_- \xrightarrow{r_+ - r_-} F_\eta].$$

For the retained theta diagram $\mathcal{T} = (V, V, \mathcal{B}, 0)$ the restrictions are Θ and $\Theta\mathcal{F}$, so its first cohomology is $Q = \mathcal{B}/\Theta V$. Tensoring the pointwise resolution, with the original signs $(-1)^{\sum_{j<i} |x_j|}$, gives top cohomology $Q^{\otimes k}$. Its numerator is $\mathcal{B}^{\otimes k}$ and its boundary subspace is the sum of tensors having a factor ΘV ; tensoring the quotient maps proves that identification. The right adjoint is the full injective complex $K_\tau(W) = [I_\eta W \rightarrow I_+ W \oplus I_- W]$, in degrees $-1, 0$, with the two signed restrictions. The complete Tau Base proof and (DB1)–(DB13) give the adjunction, tensor signs and residue-dual arrows, including the original full Taylor units.

For a full actual packet h of $g = 2\xi$, keep $E_h = \mathbb{C}[s]/(h)$, $v_h = j_h(g/h) \in E_h^\times$, and its original section $\sigma_h : E_h \hookrightarrow Q$. For the cyclic tensor-sum quotient $E = \mathbb{C}[S]/(\chi_{h,k})$, the two separately typed injections are

$$\eta_{h,k}[P] = v_h^{\otimes k} P(A_h^{(k)})1 \quad \text{in } E_h^{\otimes k}, \quad \sigma_h^{\otimes k} \eta_{h,k} : E \hookrightarrow Q^{\otimes k}.$$

Their injectivity follows from the defining cyclic annihilator, the invertible full tensor unit and the tensor of the original left inverse of σ_h . These are the original arithmetic classes in the pushforward over tau. Restriction to σ is a separate cochain map from that pushforward, defined by its original restrictions; its homotopy and fibre are retained in the full original and transcript-continuation proofs.

Result R68: the arithmetic cyclic image inside the full period family. For $d = \deg h$, the original one-factor exponential family over $\mathbb{C}[u, t]$ is $[\mathbb{C}[u, t][s] \xrightarrow{u\partial_s + h(s) - t} \mathbb{C}[u, t][s] ds]$. Use the original primitive $\Phi'_h = h$, its fixed additive constant, and the oriented rapid-decay contours in the admitted sector of $u \neq 0$. Their full matrix $\Pi_h(u, t)$ has columns $\int_{\Gamma_j} s^a e^{(\Phi_h(s) - ts)/u} ds$, $0 \leq a < d$. The complete period proof retains its determinant, ray phases, contour orientations and all coefficients. With independent parameters $t_1, \dots, t_k$, the unchanged rectangular matrix

$$\mathcal{P}(u, \mathbf{t}) = \left(\bigotimes_{i=1}^k \Pi_h(u, t_i) \right) \eta_{h,k}$$

is injective, because each period factor is invertible and $\eta_{h,k}$ is injective. The exact derivative at the undeformed parameter is

$$-u \sum_{i=1}^k \partial_{t_i} \mathcal{P}(u, \mathbf{0}) = \mathcal{P}(u, \mathbf{0}) A_{h,k}.$$

Differentiating each exponential inserts s_i ; the cyclic intertwining and commutation of the full tensor unit with $\sum A_i$ prove the formula. It uses the original sum coordinate, with no division by k . The full (DP23)–(DP33) proof supplies the coefficient-ring reductions, exact left inverse on this image, original metric pullback and separately defined finite-field external product. All nonsemisimple jets and root collisions have their stated domains there.

Result R69: actual theta dilation and the moving-parameter flow. On the original theta source use $U_a F(x) = F(x/a)$ for positive real a ; its Mellin action is a^s . The two-leg action is $(\phi, \psi) \mapsto (U_a \phi, a U_{1/a} \psi)$. The complete scaling proof computes the original source boundary of its packet section and the finite cocycle, so the quotient section intertwines this action with the full exponential of the original jet operator, including all nilpotents. For the h exponential family the regular connection operator is

$$L = u\partial_t - s, \quad [u\partial_s + h - t, L] = 0, \quad \frac{dC_b}{db} = -C_b A_h(t + ub), \quad C_0 = I.$$

The two nonzero commutators are $-u$ and $+u$. The finite matrix ODE defines the entire transport; on the admitted period domain it is exactly $\Pi_h(u, t)^{-1} \Pi_h(u, t + ub)$. Its special-fibre matrix is $C_b(0, t) = e^{-b A_h(t)}$. At

$b = \log p$ this gives the original p^{-A_h} , corresponding to $U_{1/p}$, for every positive real p . The separate global algebraic curve power map $z \mapsto z^p$ has the additional domain $p \in \mathbb{Z}_{>0}$; there its pullback is $dz/z \mapsto p dz/z$ and the divisor depth is multiplied by that integer. The corrected full source states these distinct parameter domains explicitly.

For the retained reflection-stable cyclic polynomial, put $\varepsilon = (-1)^q$, $q = \deg \chi$, and $RP(S) = P(k - S)$. The full reflection proof gives

$$RA_\chi(t)R = kI - A_\chi(\varepsilon t), \quad RC_b^\chi(u, t)R = e^{-kb}C_{-b}^\chi(-\varepsilon u, \varepsilon t).$$

Here the cyclic potential is $\Phi'_\chi = \chi$, with the original endpoint $\Phi_\chi(0) = 0$. Its reflection gauge is $\exp((\Phi_\chi(k) - kt)/u)R$ on $u \neq 0$; the unsquared contour sign remains. The matrix identity extends through $u = 0$ by the entire ODE, rather than by evaluating that singular gauge. The same proof retains the inverse-transpose dual and the original moment-line factor e^{-kb} .

Result R70: the original Taylor unit is an exact formal gauge. Let ν be the degree-less-than- d polynomial representing the full v_h , so $\gcd(h, \nu) = 1$. The exact rings are $\widehat{B} = \mathbb{C}[s][[u, t]]$ and $\widehat{B}_\nu = \mathbb{C}[s, \nu^{-1}][[u, t]]$. The coefficientwise pole quotient is $M[[u, t]]$, $M = \mathbb{C}[s, \nu^{-1}]/\mathbb{C}[s]$. Multiplication by h on M has an exact inverse H : Bezout coefficients for h, ν^n prove surjectivity on each principal part and coprimality proves injectivity. With $E_0 = u\partial_s - t$ the inverse on the pole complex is

$$J = \sum_{n \geq 0} (-HE_0)^n H.$$

The displayed order is essential. Each factor HE_0 raises parameter order, making every coefficient a finite sum. The proper-rational section of the quotient gives the explicit contraction and retraction in (UG8)–(UG11). Multiplication by the full ν then gives

$$\nu(u\partial_s + h - t)\nu^{-1} = u\partial_s + h - t - u\nu'/\nu.$$

It commutes with the same $L = u\partial_t - s$. Its endpoint on E_h is exactly multiplication by v_h ; the signed completed external product recovers exactly $\eta_{h,k}$ after the original cyclic inclusion. The complete (UG1)–(UG19) proof includes every tensor homotopy and the inverse-unit residue dual on its original reflection-stable domain.

Result R71: analytic pole residues, their connection and their endpoint. Return to the original h -potential $\Phi'_h = h$ with its specified additive constant. For fixed $u \neq 0$, write $w = e^{(\Phi_h(s) - ts)/u}$ and let $\mathcal{P}_\nu$ be the finite set of distinct zeros of ν , of size r . The exact pole sequence now is

$$0 \longrightarrow M \xrightarrow{u\partial_s + h - t} M \xrightarrow{F \mapsto (\text{Res}_p(wF ds))_p} \mathbb{C}^{\mathcal{P}_\nu} \longrightarrow 0.$$

The highest pole term $c(s - p)^{-n}$ maps to $-unc(s - p)^{-n-1}$, proving injectivity. A zero-residue Laurent series wF has a meromorphic primitive without a logarithm; taking the principal part of that primitive divided by uw proves exactness in the middle. The lift $1/(w(p)(s - p))$ proves surjectivity. The original polynomial cohomology therefore injects into the analytic localization, with precisely this residue cokernel. A full basis consists of the original d polynomial classes and the simple-pole classes $1/(s - p)$.

Let $P = \text{diag}(p)$ and let B_0 have each column the original constant-polynomial vector e_0 . Their exact connection and transport are

$$L = u\partial_t - \begin{pmatrix} A_h(t) & B_0 \\ 0 & P \end{pmatrix}, \quad C_{\text{ext},b} = \begin{pmatrix} C_b & K_b \\ 0 & e^{-bP} \end{pmatrix}, \quad K_b = - \int_0^b C_c B_0 e^{-(b-c)P} dc.$$

The off-diagonal term follows from $L(1/(s - p)) = -p/(s - p) - 1$ and the displayed ODE. The full routed period determinant is $\det \Pi_{\text{ext}} = \det \Pi_h (2\pi i)^r \prod_p w(p)$, with positive small-loop orientations. Under the actual unit gauge the weight is w/ν ; all periods agree because $(w/\nu)(\nu F) = wF$.

At the original formal endpoint let Q_0 have columns $-[(h(s) - h(p))/(s - p)h(p)]$ in E_h . Then

$$A_h Q_0 = B_0 + Q_0 P, \quad J_0 = [I \ Q_0], \quad \ker J_0 = \{(-Q_0 c, c) : c \in \mathbb{C}^r\}, \quad J_0 C_{\text{ext},b}(0, 0) = e^{-bA_h} J_0.$$

Multiplication by the full v_h after J_0 is the gauge endpoint. These are explicit comparison maps between the analytic extension and the original formal fibre. The complete (AP1)–(AP26) proof retains all higher poles, the ordinary dual, the zero-pole case and the exceptional $p = 0$ phase; it proves no evaluation of the formal inverse at a nonzero complex u .

Result R72: the signed arithmetic determinant variation. On the unchanged common source $\mathcal{P}_{2q}$ interpolate the original Gamma and arithmetic densities as $m_t = (1-t)m_\Gamma + tm_{\text{ar}}$, and write M_N, H_N, R_N, G_N for the original source Gram, relation Gram, minimum section and quotient Gram. The complete (AT1)–(AT22) proof preserves the determinant-line sign and gives

$$\begin{aligned} V_N &= \det G_N = \det M_N / \det H_N, & \dot{R}_N &= -B_N H_N^{-1} B_N^* \Delta M_N R_N, \\ \dot{G}_N &= R_N^* \Delta M_N R_N, & \ddot{G}_N &= -2R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N. \end{aligned}$$

Both negative squared norms in the logarithmic second derivative remain. The full (AV1)–(AV32) calculation proves that equality holds exactly when $\Delta M_N R_N = 0$. With P_N, Q_N the original same-metric polynomial and relation projections, retain

$$\begin{aligned} U &= Q_{2q-1} + Q_{2q} - Q_q, & W &= P_{2q-1} - P_{q-1} + P_{2q} - P_q, \\ C(t) &= M_{2q}(t)^{-1} \Delta M_{2q}, & \mathcal{B}(t) &= \log \frac{V_{q-1}(t) V_q(t)}{V_{2q-1}(t) V_{2q}(t)}. \end{aligned}$$

Subtracting the four exact determinant derivatives proves

$$\mathcal{B}(1) - \mathcal{B}(0) = \int_0^1 \text{Tr}((U - W)C) dt.$$

The full proof also gives its absolutely convergent pointwise kernel integral against $m_{\text{ar}} - m_\Gamma$, including zeros of the arithmetic density. It assumes no bounded pointwise ratio of the two densities.

Result R73: full spectral data and the exact source isometry. The nested original flags give both U and W the full spectrum $0^{[q]}, 1^{[2]}, 2^{[q-1]}$. The explicit original-metric isometry T in (AW9)–(AW10) intertwines them and gives

$$U = TWT^{-1}, \quad \text{Tr}((U - W)C) = \text{Tr}(WT^{-1}[C, T]).$$

Its original arithmetic observation is explicitly $\sigma_h^{\otimes k} \eta J_{2q} T$, with defect $\sigma_h^{\otimes k} \eta J_{2q} (T - I)$ and kernel $\mathcal{P}_{q-1}^\perp$. Thus its action on original classes is retained rather than presumed to be identity. For all eigenvalues $b_1 \leq \dots \leq b_{2q+1}$ of the actual relative source Gram, counted with multiplicity, the complete spectral proof gives

$$|\mathcal{B}(1) - \mathcal{B}(0)| \leq \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log \frac{b_{2q+1}}{b_1}.$$

The full (FS1)–(FS35) proof gives strict pointwise inequality for nonscalar C on the literal polynomial flags, and an exact limiting positive-coefficient-metric family. Its moment obstruction is explicitly nonzero; the constructed limiting family is not assigned an arithmetic density. The complete (AW1)–(AW23), (WS1)–(WS39) and independent spectral proofs retain the precise equality scope and all relative eigenvalues.

Result R74: all-degree nonlinear control from the original moments. The inherited endpoint-window proof already gives the canonical principal-angle comparison, nonlinear arcosh transport, strict moment positivity and signed tensor theta primitive (EW.25)–(EW.28), (EW.35a)–(EW.35c), (EW.46a)–(EW.48). The complete (AGT1)–(AGT19) and (AA1)–(AA44) proofs independently rederive these structures and sharpen the selected window $n = r = q$: crossing its same four projection terms changes the guaranteed moving-angle count from $2q$ to $2q - 1$. The full comparison (EP1)–(EP36) proves every coefficient, metric and operator identification and retains the inherited proof's broader window domain. For two nested cutoffs the actual relation-valued difference of minimum sections satisfies

$$D_{ij} = R_i - R_j, \quad G_i - G_j = D_{ij}^* M D_{ij}, \quad K_{ij} = G_i^{-1/2} G_j G_i^{-1/2}.$$

The eigenvalues of K_{ij} are the squared principal cosines of the original minimum-section subspaces, with their full multiplicities. The pairs $(q-1, 2q)$ and $(q, 2q-1)$ give the original $\mathcal{B}$ as their combined negative log determinant. The second pair has at least one exact zero angle in an ambient space of dimension $2q-1$. Retaining the full angle lists thus leaves $m = 2q-1$ possibly nonzero contributions.

For $\chi(S) = \sum_{j=0}^q a_j S^j$ define $\chi^{\#k}(S) = \sum_{j=0}^q \bar{a}_j (k-S)^j$. The actual line $\bar{S} = k - S$ gives $\chi^{\#k}(S) = \overline{\chi(S)}$ there, with $\chi^{\#k} \in \mathcal{P}_q$. If $\mathcal{B}$ vanished, its first pair would coincide, forcing $1 \perp \chi \mathcal{P}_q$ and hence $0 = \int |\chi(k/2 + iu)|^2 m_t(u) du > 0$. Thus $\mathcal{B}(t) > 0$ throughout the actual interpolation. The exact projection-difference spectra and concavity of $\sqrt{1 - e^{-x}}$ now yield

$$|\dot{\mathcal{B}}| \leq m \sqrt{1 - e^{-\mathcal{B}/m}} (c_{\max} - c_{\min}), \quad |2 \text{arcosh}(e^{\mathcal{B}(1)/(2m)}) - 2 \text{arcosh}(e^{\mathcal{B}(0)/(2m)})| \leq \log \kappa,$$

where $\kappa = b_{2q+1}/b_1$ is the original finite relative Gram ratio. The proof includes the explicit hyperbolic-cosine endpoint interval, the $q = 1$ case, every zero angle, the original theta primitive of D_{ij} and its full source energy. It applies in particular to the original full quartet value $q = [1 + k(m_h - 1)](k + 1)^2$, where m_h denotes the quartet's common zero order; this last m_h is distinct from the angle count $2q - 1$.

For the original arbitrary window $n \geq q$, $r \geq 1$, the same crossed pairing $(n - 1, n + r)$ and $(n, n + r - 1)$ gives the proved count $a(q, r) = \min(q, r + 1) + \min(q, r - 1)$ in (EP17)–(EP25). It improves the inherited count $2 \min(q, r)$ by one exactly when $r = q$. Its entire loss, norm factor, full Taylor units and supported theta primitive retain the original definitions.

The complete proofs following these results retain the original arithmetic Grams as actual data. Their uniform control in tensor degree is the continuing analytic calculation; no missing estimate has been introduced as an assumption in the bounds just proved.

Complete independent crosswalk of R67–R74

Independent mathematical crosswalk: current Results R67–R74

Status: accepted after complete formula review, source-domain repairs and resolved EW attribution.

Current fragment SHA-256: b87d27f38f882a80686de2a15fd85362dbd3e4692fbd74200a3b18bf8efd0171.

The final fragment was read completely after the three reported repairs and again after its EW attribution and general-window addition. All findings and the relative-source attribution are resolved; the final exact pin is accepted.

13 September 2026. This review concerns the complete current-results fragment and its stated original-source comparisons. It is not a replacement proof of collateral literature results, an audit of the whole cumulative archive, or a claim of an arithmetic bound uniform in tensor degree.

R67 — actual τ pushforward and original arithmetic injections

The original Tau Base note, equations (17)–(22), and DP/DB review DB4–7 use the covariant coefficient-diagram category on the literal four-point poset. Evaluation identifies $\text{Hom}(P_x, F)$ with F_x . At each lower point the augmentation of $P_+ \oplus P_-$ is the identity, and at each upper point its kernel consists exactly of $(v, -v)$. Thus the projective resolution and the resulting differential $r_+ - r_-$ in the fragment agree pointwise with the source. No restriction to the σ stalk has replaced this derived global-section map.

The tensor numerator is the original tensor of copies of the target space B . Its boundary is the sum of tensors with a θ -image factor. Taking the tensor of the quotient maps over $\mathbb{C}$ therefore identifies its quotient with $Q^{\otimes k}$, with the stated Koszul differential signs. The dual complex has degrees -1 and 0 and signed restrictions $(+ \text{res}, - \text{res})$, exactly source (27)–(30) and DB8–9. The fragment does not assign an unproved positive phase to the raw residue form or insert an explicit tensor-Hom coefficient convention contrary to the full comparison supplied later.

For the actual full-order packet, DP23–27 and DB10–11 give two distinct arrows. Polynomial evaluation modulo the cyclic annihilator injects E into $E_h^{\otimes k}$. Multiplication by the full invertible unit $v_h^{\otimes k}$ preserves injectivity and commutes with the sum operator. Tensoring the actual section σ_h then injects this image into $Q^{\otimes k}$. The original jet map is a left inverse of σ_h , so the last injectivity assertion has an exact map proving it. These codomains are retained correctly in R67. The separate restriction to σ and its homotopy are source (20), with homotopy $r_{\eta, \sigma}$; the fragment does not infer vanishing of global classes from that map.

R68 — coefficient family and rectangular period inclusion

The family coefficient ring must be $\mathbb{C}[u, t][s]$, as in DP2–4; $\mathbb{C}[s]$ denotes only a fibre after fixing numerical parameters. This distinction was reported to the author because the first reviewed fragment displayed $\mathbb{C}[s]$ while calling the object the family. The degree-one module is polynomial one-forms, with the factor ds part of its type.

DP23–33 proves the full claimed construction: the one-factor remainder module has rank $d = \deg h$, and its external tensor product over the independent-parameter ring has top cohomology $E_h^{\otimes k}$ at the undeformed point. The full period matrix has entries $\Pi_{j,a} = \int_{\Gamma_j} s^a \exp\left(\frac{\Phi_h(s) - ts}{u}\right) ds$. Its source columns are indexed by a and its target rows by oriented contours j . In the fixed admitted nonzero- u sector every period factor is invertible, so their tensor multiplied on the right by η has the asserted injective rectangular map and original domain E .

Differentiation with respect to t_i contributes multiplication by $-s_i/u$. At all $t_i = 0$ the companion operator is the original A_h in each factor. Therefore $-\sum_i \partial_{t_i} P = (\otimes \Pi_h) A_h^{(k)} \eta = P A_{h,k}$, using the actual cyclic intertwining. This proves the exact order and the coefficient one of the sum; no averaging factor k is introduced. DP31 states the derivative only at the undeformed parameter. R68 keeps that scope and does not presume invariance of the rectangular image under every deformed operator. DP32 retains the actual quotient metric pullback, and DP33 separately supplies the finite-field external product and its stated reduction domain.

R69 — arithmetic dilation, moving transport and reflection

The full scaling source Sections 2–5 and the original Tau Base equations (32)–(37) show that $U_a F(x) = F(x/a)$ acts by a^s under Mellin transform. The two source-leg operators are U_a and $aU_{1/a}$, because $\mathcal{F}U_a = aU_{1/a}\mathcal{F}$.

Both the differential and all support maps intertwine this action. The finite source correction is a literal θ boundary with its integral primitive, hence the induced quotient section is equivariant, including all nilpotent jet coefficients.

For $D = u\partial_s + h - t$ and $L = u\partial_t - s$, the only nonzero commutators in $[D, L]$ are $-u$ and $+u$. This gives $[D, L] = 0$. The finite transport is the right-ordered ODE $C'_b = -C_b A_h(t + ub)$, $C_0 = I$. The retained period equation $\partial_t \Pi_h = -\Pi_h A_h(t)/u$ proves $C_b = \Pi_h(u, t)^{-1} \Pi_h(u, t + ub)$, with matrix multiplication in precisely that order. At $u = 0$ its specialization is $\exp(-bA_h(t))$; $b = \log p$ gives p^{-A_h} at $t = 0$ for every real $p > 0$. The separately stated algebraic $z \mapsto z^p$ map requires p a positive integer. R69 correctly retains that repaired domain and does not attach it to all real θ dilations.

The reflection calculation is for the cyclic χ -potential, not the h -potential from R68. In RW4 its retained primitive satisfies $\Phi'_\chi = \chi$ and $\Phi_\chi(0) = 0$. This required naming repair was reported to the author. The polynomial identity $\chi(k - S) = \epsilon\chi(S)$, $\epsilon = (-1)^q$, yields $RA_\chi(t)R = kI - A_\chi(\epsilon t)$. The parameter map is $(u, t) \mapsto (-\epsilon u, \epsilon t)$, and the horizontal gauge is $\exp\left(\frac{\Phi_\chi(k) - kt}{u}\right) R$. Its logarithmic t -derivative is $-k/u$, canceling the exact connection defect. Its contour substitution retains the one-form minus sign in RW13. Applying the ODE and its initial condition gives $RC_b R = \exp(-kb)C_{-b}(-\epsilon u, \epsilon t)$, for all parameters by the entire matrix solution. This last identity extends through $u = 0$ without evaluating the singular contour gauge there. RW19–21 retains the inverse-transpose and the original $\exp(-kb)$ moment-line factor.

R70 — full Taylor-unit formal gauge and contraction

UG1–19 works on the original degree-below- d polynomial ν representing the entire Taylor unit, not only its constant coefficient. Full selected zero orders give $\gcd(h, \nu) = 1$. On $M = \mathbb{C}[s, \nu^{-1}]/\mathbb{C}[s]$, multiplication by h is injective by coprimality and surjective by Bezout against every ν^n . Its inverse H is defined on this pole quotient; it is not an inverse to h on E_h .

The coefficientwise completed rings and quotient in R70 agree with UG4–5. With $E_0 = u\partial_s - t$, $D_M = M_h(I + HE_0)$. The ordered series $J = \sum_{n \geq 0} (-HE_0)^n H$ is therefore the two-sided inverse. Each added factor raises the (u, t) -adic order, so each coefficient is a finite sum; no commutation of H and ∂_s is used. The proper-rational section yields the explicit degree -1 homotopy and retraction in UG10–11. R70 correctly scopes this inverse to the formal completion.

Directly differentiating P/ν proves $\nu D\nu^{-1} = u\partial_s + h - t - u\nu'/\nu$. Multiplication by ν is the cochain isomorphism, and $[L, \nu] = [L, \nu^{-1}] = 0$ proves that the same L intertwines it. At the original fibre the map is exactly multiplication by ν_h . UG18 tensors that full unit in the specified completed coefficient space; composing with the existing cyclic inclusion gives η . UG19 keeps the inverse Taylor unit and reflection signs in the residue-dual map. The fragment retains these maps and does not evaluate the formal Neumann series at nonzero u .

R71 — analytic pole sequence, extension transport and exact endpoint

AP1–26 fixes u in $\mathbb{C}^*$ and t in $\mathbb{C}$, with the original h -potential Φ_h and its original additive constant. The residue map is $\text{Res}_p(wF ds)$, so ds must remain inside its residue argument. The first reviewed fragment had a literal comma in this expression; the differential-notation repair was reported.

In M , a leading principal part $c(s - p)^{-n}$ maps under D to $-unc(s - p)^{-n-1}$. The multiplicative part has order at most n , so injectivity follows without canceling any pole. A zero-residue Laurent series wF has a meromorphic primitive; summing the principal parts of that primitive divided by uw supplies Q with $DQ - F$ polynomial. This proves the exact middle kernel. The lift $\frac{1}{w(p)(s-p)}$ proves surjectivity with its exact nonzero factor. The long exact sequence, or the same leading-pole argument, injects the polynomial cohomology and leaves exactly the residue cokernel. The d original monomial classes and r simple poles form the full basis, as AP5–6 proves.

On the simple-pole basis, $L\left(\frac{1}{s-p}\right) = -\frac{p}{s-p} - 1$. Therefore the block companion is $\begin{bmatrix} A_h(t) & B_0 \\ 0 & P \end{bmatrix}$, with each B_0 column e_0 , and the connection is $u\partial_t$ minus that matrix. Differentiating $K_b = -\int_0^b C_c B_0 \exp(-(b-c)P) dc$ yields $K'_b = -C_b B_0 - K_b P$. This is exactly the upper-right block of the right-ordered transport ODE. The full determinant is $\det \Pi_h (2\pi i)^r \prod_p w(p)$; it uses the routed rapid-decay contours followed by positively oriented small loops. Under multiplication by the actual unit, the target weight w/ν cancels ν only in the identity $(w/\nu)(\nu F) = wF$; no source norm has been replaced.

The endpoint polynomial $Q_{0,p} = -\frac{h(s)-h(p)}{(s-p)h(p)}$ represents $(s-p)^{-1}$ modulo h because $h(p)$ is nonzero. Thus $A_h Q_0 = B_0 + Q_0 P$. Setting $J_0 = [I, Q_0]$ gives $J_0 A_{\text{ext}}(0) = A_h J_0$ and the exact kernel $\{(-Q_0 c, c)\}$. Matrix

exponentiation then gives $J_0 C_{\text{ext},b}(0,0) = \exp(-bA_h)J_0$. This is a matrix-specialization comparison to the rank- d formal fibre; it does not identify the analytic rank- $-(d+r)$ cohomology with that fibre. The full gauge endpoint is multiplication by v_h after J_0 . The stated $p=0$ and $r=0$ scopes are present in AP and retained by the fragment.

R72 — signed determinant derivative and curvature equality

AT3–10 and AV1–32 keep B_N equal to literal multiplication by χ and J_N literal remainder. Orthogonality gives $R_N = s_N - B_N H_N^{-1} B_N^* M_N s_N$, $J_N R_N = I$ and $B_N^* M_N R_N = 0$. The decomposition $p = B_N y + R_N J_N p$ implies $M_N R_N = J_N^* G_N$. This proves the inverse quotient Gram without assuming it. The determinant-line sign is $(-1)^{q(N-q+1)}$; its squared modulus in the metric identity gives $V_N = \frac{\det M_N}{\det H_N}$.

Differentiating $J_N R_N = I$ shows that the derivative of R_N is relation-valued. Differentiating its orthogonality determines $\dot{R}_N = -B_N H_N^{-1} B_N^* \Delta M_N R_N$. The two derivatives of G_N retain both terms and give precisely the factor -2 in R72. The logarithmic second derivative is minus twice the first Hilbert–Schmidt square and minus the second square. Equality forces $B_N^* \Delta M_N R_N = 0$ and $R_N^* \Delta M_N R_N = 0$; invertibility of $[B_N, R_N]$ then gives $\Delta M_N R_N = 0$. Conversely this equation kills both squares. This includes the empty relation domain at $N = q-1$.

For every cutoff, the same-metric projection formula gives $(\log V_N)' = \text{Tr}((P_N - Q_N)C)$. Since $Q_{q-1} = 0$, the fragment's $U = Q_{2q-1} + Q_{2q} - Q_q$ is exactly AT13, and its W has all four original degree projections. Subtracting the two larger endpoint derivatives from the two smaller ones gives $\text{Tr}((U - W)C)$, with the displayed sign. The integral needs the differential dt rather than a comma. AT16–17 supplies the full pointwise kernel: its coefficient matrices are bounded on the compact t interval and $|\Delta m| \leq m_0 + m_1$ with all finite moments. This proves absolute integrability even at zeros of the arithmetic density; no bounded density ratio is used.

R73 — all eigenvalues, source isometry and strictness scope

The nested relation flag has dimensions $1, q, q+1$. U acts by $1, 2, 1, 0$ on its successive orthogonal pieces of dimensions $1, q-1, 1, q$. The degree flag gives W actions $0, 1, 2, 1$ with dimensions $q, 1, q-1, 1$. This proves their common full spectrum $0^{[q]}, 1^{[2]}, 2^{[q-1]}$, including $q=1$. AW9–10 maps the ordered orthonormal polynomial basis to the original relation basis and canonical quotient basis in the same metric; thus $T^*MT = M$ and $U = TWT^{-1}$. Trace cyclicity gives the displayed commutator expression.

The arithmetic observer $O = \sigma_h^{\otimes k} \eta J_{2q}$ has kernel $\chi \mathcal{P}_q$. AW10 maps $\mathcal{P}_{q-1}^\perp$ exactly onto that relation space. Consequently $\ker(OT) = \mathcal{P}_{q-1}^\perp$, and the actual defect is $O(T - I)$. This is the kernel of the observation after T , not an assertion that the defect has that kernel. All original units and quotient arrows survive this calculation.

For C with ordered eigenvalues c_1 through c_{2q+1} , the rank- $-(q+1)$ and rank- $-(q-1)$ spectral projections of U and W give width $c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j)$. For the actual affine Gram path these eigenvalues are $\frac{b_j-1}{1-t+tb_j}$, in the same order for every t ; integrating gives $\log b_j$. Therefore the first displayed bound and the coefficient $1 + 2(q-1) = 2q-1$ of $\log \kappa$ have the exact stated indices and multiplicities.

FS14–19 proves strictness for every nonscalar C using the literal incidences $\mathcal{P}_{q-1} \cap \chi \mathcal{P}_q = 0$ and $\mathcal{P}_q + \chi \mathcal{P}_q = \mathcal{P}_{2q}$. Equality of any nontrivial-rank extremal trace would contradict these incidences. The spectral-gap expansion then gives strictness even with repeated eigenvalues. FS20–32 constructs the exact positive coefficient-metric family approaching the width, with trace $\frac{B_q(c)}{1+\epsilon^2}$; the endpoint $\epsilon=0$ is singular. FS33–35 computes its failure of the actual moment identity as μ for $q \geq 2$ or 2μ for $q=1$. Thus the limiting family is not claimed as an arithmetic density. R73 retains both scopes. The separate WS1–39 source has the asserted tag range and is included as another full proof body.

One exposition clarification in the underlying FS compactness paragraph is retained explicitly here. Its equation $E^* = E$ uses an M -orthonormal frame. In the unchanged polynomial frame the condition is $E^* M = M E$. The exact map $E \mapsto M^{1/2} E M^{-1/2}$ carries these M -orthogonal idempotents with trace r to ordinary Hermitian idempotents with trace r ; its inverse is $F \mapsto M^{-1/2} F M^{1/2}$. Both maps are continuous, since M is fixed positive definite. The ordinary projection set is closed and bounded, hence compact, so the original M -projection set is compact. This supplies the needed precise coordinate map for FS17–19 and changes no mathematical inequality. The separate FS/AV reviewer also checked both proof bodies in full and recorded this clarification.

R74 — all-degree relation angles and nonlinear estimate

The final attribution was checked against the complete EP1–36 comparison and against the original EW source directly. EW25–28 already has the canonical projection $R_j = \Pi_j R_i$, the full relative-Gram cosine list, every

common direction and the gap bound $\min(q, j - i)$. EW35a–c already derives the nonlinear arcosh map by concavity and integrates it, including zero losses for general positive matrix paths. EW46a already proves strict positivity for every nonempty actual positive moment window, using the original monic three-term recurrence; EW46–48 already constructs the signed tensor θ primitive by successive division with the same full packet and seed. The final opening of R74 attributes all these inherited statements correctly. Its comparison concerns these pinned bodies; it does not assign an independently verified external priority date.

EP1–9 preserves the exact maps needed for that attribution. The remainder-coordinate map from EW's C to AGT's E is the identity, and EW's source Gram H_N is AGT's M_N after the original inclusions. At $\lambda = 1/4$, $c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda) = \sqrt{2\pi}$, so the reference density, its mass $(2\pi)^{k/2}$, and the arithmetic mass μ_h^k coincide literally. EW's contravariant minimum-section matrix Q_N is converted to AGT's projection by $Q_N M$; it is not equated with AGT's relation projection. This gives $\text{Tr}(Q_N M) = \text{Tr}(\Pi_N C)$ with $C = M^{-1}M$. Consequently the same four signed derivative terms and every original determinant ratio enter the comparison.

For $i \leq j$, $J_j(R_i - R_j) = 0$ and R_j is orthogonal to that original relation space. Hence $R_j = Z_j R_i$ and $G_i - G_j = (R_i - R_j)^* M (R_i - R_j)$, with each inclusion into the common source retained. The isometries $R_i G_i^{-1/2}$ and $R_j G_j^{-1/2}$ have cross matrix $G_j^{1/2} G_i^{-1/2}$. Its squared singular values are exactly the eigenvalues of $G_i^{-1/2} G_j G_i^{-1/2}$, all in $(0, 1]$. These are the full squared principal cosines. Their product gives V_j/V_i without commuting either Gram.

The two pairs in R74 reproduce precisely the original four-volume. For the second pair, both q -dimensional subspaces lie in $\mathcal{P}_{2q-1} \cap (\chi \mathcal{P}_0)^\perp$, of dimension $2q - 1$; therefore one angle is exactly zero. The two full lists retain every eigenspace while only the scalar list has $2q - 1$ possibly nonzero contributions. At $q = 1$ the second pair is identical. This count is unrelated to the original quartet zero multiplicity m_h .

On the original line $S = k/2 + iu$, $\bar{S} = k - S$ proves the displayed polynomial involution identity. If the four-volume were zero, all nonnegative costs of its first angle pair would vanish. That would identify $\mathcal{P}_{q-1}$ with $(\chi \mathcal{P}_q)^\perp$, making 1 orthogonal to $\chi \chi_k^\#$. Its inner product is the strictly positive integral of $|\chi|^2$ against the actual density, a contradiction. The differential du must be written as part of this integral. No arbitrary positive coefficient form supplies this moment argument.

Each nonzero angle contributes eigenvalues plus and minus $\sin \theta$ to a projection difference. Pairing those eigenprojections with C bounds the absolute derivative by the spectral width times the sum of $\sin \theta$. For $x = -\log \cos^2 \theta$, $\sin \theta = \sqrt{1 - \exp(-x)}$, a concave function on $[0, \infty)$. The exact sum of the x values is B , so the retained tangent inequality bounds their sum by $m\sqrt{1 - \exp(-B/m)}$, $m = 2q - 1$. The positive derivative of $2 \text{arcosh}(\exp(B/(2m)))$ is $\frac{1}{m\sqrt{1 - \exp(-B/m)}}$. Strict positivity of B on the compact path makes this division valid throughout. Integrating the original relative spectral width gives $\log \kappa$, proving the final bound with no missing factor m . AGT19 provides its explicit inverse interval. All θ relation primitives and their nonnegative original energies remain in AGT5 and the source comparison following it.

The added general-window paragraph also checks in its full domain $n \geq q \geq 1$, $r \geq 1$. Pairing $(n - 1, n + r)$ and $(n, n + r - 1)$ retains the original four-volume because the product of the two quotient ratios is exactly $\frac{V_{n-1}V_n}{V_{n+r-1}V_{n+r}}$. The corresponding difference of projections has the same four terms and signs. Their gaps are $r + 1$ and $r - 1$, so the inherited rank theorem bounds their respective moving-angle counts by $\min(q, r + 1)$ and $\min(q, r - 1)$. At $r = 1$ the second pair is identical and its entire angle list consists of zeros. Thus $a(q, r) = \min(q, r + 1) + \min(q, r - 1)$ is always at least one and accounts for every possible moving slot.

For $r < q$, integrality gives $r + 1 \leq q$ and $a = (r + 1) + (r - 1) = 2r$. For $r = q$, $a = q + (q - 1) = 2q - 1$. For $r > q$, integrality gives $r - 1 \geq q$ and $a = 2q$. These three exhaustive cases prove that the decrease from $2 \min(q, r)$ is exactly one when $r = q$ and zero otherwise; $q = r = 1$ gives $a = 1$, while $q = 1 < r$ gives $a = 2$. The count does not replace the ambient cutoff $n + r$ or the original multiplicities.

For completeness the general inequality follows from the same actual pairs: let L be the unchanged logarithmic four-volume and use all their nonzero costs, appending zero scalar terms up to a slots. Concavity gives $|L'| \leq a\sqrt{1 - \exp(-L/a)}$ times the original spectral width. The derivative of $\Psi_a(L) = 2 \text{arcosh}(\exp(L/(2a)))$ is $\frac{1}{a\sqrt{1 - \exp(-L/a)}}$. At possible zero losses the regularized derivative of $\Psi_a(L + \epsilon)$ is bounded by the width times $\sqrt{\frac{1 - \exp(-L/a)}{1 - \exp(-(L + \epsilon)/a)}}$ $\leq$ the width. Integration followed by ϵ decreasing to zero proves the same endpoint bound, and the affine Gram path integrates its width to $\log \kappa$. The inverse map is exactly $2a \log \cosh(v/2)$, yielding EP24–25 with exponent $2a$. EW's original source norm factor $U = \omega_{n+r}/\omega_n$ remains in its full endpoint $U^{1/(2r)} \sinh(L/(2r))$; the angle-count refinement controls L and does not delete that factor.

Finally EP32–36 identifies the actual relation polynomial, not only its class. Injectivity of multiplication by χ makes EW's coefficient κ_x equal to $X_{ij}x$ with $X_{ij} = (B_j^* M B_j)^{-1} B_j^* M R_i$. Its retained energy is $X_{ij}^* (B_j^* M B_j) X_{ij}$. Both constructions successively divide $\chi(\sum s_i)$ by the same monic $h(s_i)$, in the same order; uniqueness therefore

gives the same quotient polynomials. The primitive's coefficient $(-1)^{i-1}$ cancels precisely the tensor differential's sign from its $i - 1$ preceding degree-one factors. Its image is $V_{h,k}(\chi P)$, with the same original $h(D)F_h = \Theta\phi_0$ and the same complete unit η . The usual one-leg inclusion $\phi \mapsto (\phi, 0)$ carries it into the actual τ two-leg complex without a sign change, and every supported zero stays at its original label. These direct source checks resolve the new paragraph's preservation claims and its attribution to the inherited construction.

Scope of acceptance

The mathematical equations checked above preserve the actual original arithmetic and Gamma densities, finite cutoff spaces, full Taylor units, parameter shifts, source and target degrees, signs, and every relative eigenvalue in the stronger spectral bound. They establish finite exact and nonlinear bounds. They do not assert a bound for the relative Gram uniform in k , and they do not establish an RH result. That scope agrees with the fragment's last paragraph.

Attribution within the compared source graph is now resolved by the complete EP comparison and the direct EW reading recorded above. The final fragment correctly identifies the inherited methods and conclusions and attributes the later content to the crossed-pair count refinement and its exact general-window consequence. It does not claim that AGT/AA first supplied all-degree angles, nonlinear transport, strict moment positivity or the tensor θ primitive. No external historical priority or publication date is certified. The earlier mathematical acceptance of fragment 56c265751fca6499c62ce649da18c3aef4dfdbe7d717df032d5d0fe9b3cd92cb remains part of the review history; the final accepted pin additionally includes the now-resolved attribution and the general-window paragraph.

No Lean process, remote mutation, PDF render, numerical proof certificate, or review of unrelated archive contents was performed in this crosswalk. The attached JSON records exact source bytes and line locators, with the final fragment pin only after the author repairs and the reviewer reads it back.

Source pins and locators

- **tau_base**: source:NOTE.md; 25512 bytes; SHA-256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3. Complete body; equations 1–3, 17–43 and 46–48 govern R67 and the actual base maps.
- **arithmetic_input**: source:arithmetic_input.tex; 14644 bytes; SHA-256 a50b0fb587b644c4ea94dba45b2b423d038f2bfbef7ef188543939ba850e58d2. Complete original finite-packet and source-boundary proofs A7–13, lines 179–295; also the immediately preceding Euler inverse estimates and A14–16 were read.
- **purity**: source:deligne_bound_purity_independent_review_20260913.md; 45620 bytes; SHA-256 8f5676711115ecbe926e825a98f498af74d7685d3777608c6b1b0429af3e74f1. Complete relevant proofs DP1–4, DP23–33 and DB1–13; coefficient complex, tensor periods, original τ adjunction and full-unit injections.
- **scaling**: source:ACTUAL_TAU_SCALING_AND_EXPONENTIAL_FLOW.md; 28607 bytes; SHA-256 4e4cb66e3cbbff6379ae389ddaaad65045ac0ed380f17a87ac7c4241c45e6c08. Complete Sections 1–6 and 8–10; corrected integer domain in Section 6. Section 7 finite-field calculation is outside the formulas reviewed here.
- **reflection**: source:F1_REFLECTION_WEIGHT_TRANSPORT.md; 11835 bytes; SHA-256 111f4c3ea74fa3985352d4a29a65826ee10c883c18de20230f54a7b9f15472f2. Complete body RW1–23, including $\Phi' = \chi$ and $\Phi(0) = 0$.
- **unit_gauge**: source:ACTUAL_TAYLOR_UNIT_FORMAL_GAUGE.md; 14920 bytes; SHA-256 9ded2486160b76bf65f92d44b73cdd0c59a3ca7e5334f88682e10592d5a45b2f. Complete body UG1–19.
- **analytic_poles**: source:ANALYTIC_POLE_RESIDUE_TRANSPORT.md; 18433 bytes; SHA-256 357923aeadd35c77f2fa98598fbc4ed8bc1f7cd6ebc04760da504a0ded765a652. Complete body AP1–26, including AP19a–b and the formal endpoint.
- **determinant_transport**: source:tau_arithmetic_determinant_transport_20260913.tex; 17706 bytes; SHA-256 24cf7ce5be6a43cc656641b516185df10b2e4d7824600ece657120d1c549deae. Complete body AT1–22 and AT2a.
- **window_transport**: source:tau_relation_window_spectral_transport_20260913.tex; 16528 bytes; SHA-256 c79e76c087efd59871e3234ff44fb18964515dbd34f0318dffba6b3f1440f6ec. Complete body AW1–23.
- **all_degree_angles**: source:tau_all_degree_angle_transport_20260913.tex; 15116 bytes; SHA-256 cc0c0c008b0532c5f120d1cea76b94e2d08153c87a694ce18775a5189c72d5f8. Complete body AGT1–19.
- **determinant_variation**: source:tau_arithmetic_determinant_variation_subreview_20260913.md; 12490 bytes; SHA-256 dee20e249e980fdf0251513c3ab5d504e1a17e6d408c3e7ad69d08816a6f115e. Complete body AV1–32; also independently read by the delegated FS/AV crosswalk reviewer.
- **flag_sharpness**: source:tau_four_window_spectral_sharpness_20260913.md; 18872 bytes; SHA-256

11c26c3bcfce135e1882190af9dc3a8f1e4e6487fd282cc252fa441fd90651c. Complete body FS1–35; also independently read by the delegated FS/AV crosswalk reviewer.

- **window_review**: source:tau_relation_window_spectral_review_20260913.md; 24534 bytes; SHA-256 06e611a52b369d927b90a0dbc065283ce118402905cea046fe028c81f507d96d. All WS1–39 locators and review scope inspected; the underlying complete proofs were independently checked in AW and FS above.
- **fs_av_additional_crosswalk**: source:tau_actual_current_results_fs_av_crosswalk_20260913.md; 9911 bytes; SHA-256 8a7ad968aa935114f6ac533011a92d359159e18e8c92f25577dd1c4f26ae3dbf. Complete independent FS/AV source crosswalk; supplies the explicit metric-frame homeomorphism in the compactness argument.
- **metric_compactness_clarification**: source:tau_four_window_metric_compactness_clarification_20260913.tex; 3479 bytes; SHA-256 c6246be3924763a95f6583f359596f4a7b0d0b42e60b04f476ee5b15c29500ea. Complete additive proof of the metric-coordinate homeomorphism, projection rank, compactness and attained maximum; raw FS source unchanged.
- **ew_agt_aa_provenance**: source:tau_ew_agt_aa_provenance_comparison_20260913.md; 29127 bytes; SHA-256 bdfab521d6a504d784dfa918deb3fe7af99f470abfe581df45c9c4bc4f085d6b. Complete EP1–36 comparison, including the general-window refinement and exact inherited θ -cochain equality.
- **ew_agt_aa_receipt**: source:tau_ew_agt_aa_provenance_comparison_20260913.json; 3808 bytes; SHA-256 0b4433521b1bf1731c94622e104b8834d92df349f48ef7ed749cdf099a4b5e77. Complete source receipt read; its source hashes, exact compared scope and independent-check limitations retained.
- **inherited_ew**: source:gamma_endpoint_window_bridge.tex; 54230 bytes; SHA-256 400541cdcb35d44c0f7e68f5db1107df4fcf707d0be83b5a8a970d8ec9c73160. Original source read directly at lines 1–225 (EW1–15 and full source definitions), 370–729 (complete EW25–35c projection and nonlinear transport proofs), and 883–991 (complete EW46a–48 strict moment positivity and signed θ primitive).

Exact inherited EW, AGT and AA comparison

Exact comparison of inherited EW transport with AGT and AA

13 September 2026. This add-only proof and attribution note compares the complete inherited `gamma_endpoint_window_bridge.tex` (EW.1–EW.59, including EW.35a–c and EW.46a) with the complete sealed AGT1–AGT19 and AA1–AA44 sources. All three bodies were read. No sealed source or package is changed by this note.

The inherited EW source already proves the general canonical projection and principal-angle calculation, nonlinear arcosh transport, strict positivity for the original positive polynomial moments, and the explicit signed tensor theta primitive. The later AGT calculation explicitly sharpens the chosen window by changing the pairing of the same four projections: its guaranteed scalar angle count is $2q - 1$, whereas EW.35a–c use $2q$ at that window. The proof below establishes precisely this refinement and its attribution. Strict positivity and existence of the theta primitive are not additional results of AGT or AA relative to EW.

1. Exact source pins and reading scope

The base directory in this table is `source-workspace`.

Complete source records

Source

EW

Relative path

`output/tau_split_zero_counterfactual_reconstruction_20260913/original_programme/tex/gamma_endpoint_window_bridge.tex`

SHA-256

`400541cdcb35d44c0f7e68f5db1107df4fcf707d0be83b5a8a970d8ec9c73160`

Reading scope

Entire source, EW.1–59 and all prose, including appended-letter equations

Source

Identical inherited reader copy

Relative path

`output/tau_split_zero_counterfactual_reconstruction_20260913/original_programme/sources/reader_preparation/gamma_endpoint_window_bridge.tex`

SHA-256

`400541cdcb35d44c0f7e68f5db1107df4fcf707d0be83b5a8a970d8ec9c73160`

Reading scope

Byte equality to the fully read EW source

Source

AGT

Relative path

`work/tau_all_degree_angle_transport_20260913.tex`

SHA-256

`cc0c0c008b0532c5f120d1cea76b94e2d08153c87a694ce18775a5189c72d5f8`

Reading scope

Entire corrected AGT1–19 source

Source

AA

Relative path

work/tau_all_degree_angle_independent_review_20260913.md

SHA-256

b0634428d69f80dedd4f68766ef9e6468c390eaac86dfe58b08c21c8fb896e03

Reading scope

Entire AA1–44 proof and acceptance scope

The EW introduction identifies its input criterion as PR 23 source revision c720f40530eed2f5969dbabe94dfd3fddc0f507f. That is an internal provenance statement in the read source; this comparison does not assign an independently verified publication date or external mathematical priority to it. In particular, “additional” below means additional explicit content relative to these pinned bodies. The inherited source also attributes EW.35a–c to its independent projection review. This attribution is retained.

The rational calculations EW.49–59 and their written derivations were read. Their reported checker executions were not rerun in this comparison, and their counts are not transferred to AGT or AA. The other archived EW copies and checker programs are not silently substituted for the pinned source above.

2. Original source, parameters, metrics and maps

Use EW’s original full packet h , $g = 2\xi$, integer $k \geq 1$, original line $S = k/2 + iu$, and actual cyclic relation χ of degree $q \geq 1$. Its quotient $\mathcal{C} = \mathbb{C}[S]/(\chi)$ and AGT/AA’s E have precisely the same ordered remainder coordinates. The coefficient map

$$I_E : \mathcal{C} \longrightarrow E, \quad \sum_{j=0}^{q-1} x_j[S^j] \mapsto \sum_{j=0}^{q-1} x_j[S^j] \quad (\text{EP1})$$

is the identity algebra isomorphism, with inverse the same coefficient map. It retains all multiplicities of χ . For the ambient source specialize EW’s parameters by

$$n = q, \quad r = q, \quad m_{\text{EW}} = n + r = 2q, \quad s_{\text{EW}} = \min(q, r) = q, \quad \lambda = \frac{1}{4}. \quad (\text{EP2})$$

The symbol m_{EW} is an ambient polynomial cutoff; the symbol m in AGT and AA is instead the count $2q - 1$ in a later scalar inequality. They do not denote the same integer. The exact relation between them at EP2 is $m_{\text{AGT}} = m_{\text{EW}} - 1$, while the ambient vector-space dimension is $m_{\text{EW}} + 1 = 2q + 1$.

The gamma densities coincide literally under EP2. Indeed EW.5 gives

$$c_{1/4} = 2^{1-2/4}\Gamma(2/4) = \sqrt{2\pi}, \quad r_{1/4}^{*k}(u) = \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi} = m_0(u), \quad (\text{EP3})$$

using EW.7. The gamma mass is $(2\pi)^{k/2}$, and EW.1’s arithmetic density is AGT2/AA2’s $m_1 = w_h^{*k}$, with mass μ_h^k . Therefore the common identity coefficient map on $\mathcal{P}_{2q}$ identifies the two measures at both endpoints without a mass rescaling. Along the segment $m_t = (1-t)m_0 + tm_1$, define $M(t)$ as its Gram in the fixed monomial frame. EW’s general positive-family theorem applies with

$$H_{m_{\text{EW}}}(t) = M(t), \quad H_N(t) = \iota_N^* M(t) \iota_N, \quad \dot{H}_{m_{\text{EW}}} = \dot{M} = M(1) - M(0), \quad (\text{EP4})$$

where $\iota_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}^{2q+1}$ is the original coefficient inclusion. The scalar relation χ , the monic remainder J_N , and multiplication $B_N = \times \chi$ are fixed along this same path.

EW.11 supplies

$$K_N^{\text{EW}} = J_N H_N^{-1} J_N^*, \quad G_N = (K_N^{\text{EW}})^{-1}, \quad R_N^{\text{EW}} = H_N^{-1} J_N^* G_N, \quad R_N^{\text{AGT}} = R_N^{\text{AA}} = \iota_N R_N^{\text{EW}}. \quad (\text{EP5})$$

The last equality follows both by the displayed matrix formula and by uniqueness of the minimum section: the included EW vector has remainder x , is orthogonal to $B_N \mathcal{P}_{N-q}$, and any other lift differs by a vector in that space. The squared norm of the latter is added orthogonally, so it cannot be another minimum unless the difference is zero. Hence every G_N and $V_N = \det G_N$ is the same original quantity. In particular, EW's K_N^{EW} is the inverse quotient Gram; AGT's $K_{ij} = G_i^{-1/2} G_j G_i^{-1/2}$ is a two-cutoff relative Gram. Their exact relation is the displayed substitution of $G_N = (K_N^{\text{EW}})^{-1}$, with this factor order preserved.

There is a further operator-type distinction requiring an explicit map. EW.30 defines a contravariant matrix

$$Q_N^{\text{EW}} = R_N G_N^{-1} R_N^*, \quad \Pi_N = Q_N^{\text{EW}} M = Z_N^{\text{AGT}} = N_N^{\text{AA}}. \quad (\text{EP6})$$

Here and below R_N on the common ambient space includes ι_N . To prove EP6, compute $\Pi_N^2 = R_N G_N^{-1} (R_N^* M R_N) G_N^{-1} R_N^* M = \Pi_N$. Its adjoint for M is itself, and its range is $R_N E$. Thus it is exactly the orthogonal projection that AGT denotes by $Z_N = P_N - Q_N$ and AA by N_N . EW's Q_N^{EW} is converted to that endomorphism by right multiplication by the original M ; it is not identified with AGT's relation projection Q_N^{AGT} .

For completeness EW's P_j^{EW} is the contravariant matrix of the single monic orthogonal line. Those lines span the cutoff source, so the remaining exact conversions are

$$P_N^{\text{AGT}} = \sum_{j=0}^N P_j^{\text{EW}} M, \quad Q_N^{\text{AGT}} = P_N^{\text{AGT}} - Q_N^{\text{EW}} M. \quad (\text{EP7})$$

They follow by decomposing a polynomial in the complete monic orthogonal family and then subtracting its minimum-section component. Every factor M remains in the written map.

Set $C = M^{-1} \dot{M}$, the AGT4/AA21 operator. The map $v \mapsto M^{1/2} v$ is an isometry from $(\mathcal{P}_{2q}, M)$ to its displayed Euclidean coefficient space, and

$$M^{1/2} C M^{-1/2} = M^{-1/2} \dot{M} M^{-1/2}, \quad \mathfrak{o}_H(\dot{H}) = c_{\max}(C) - c_{\min}(C). \quad (\text{EP8})$$

The trace also has the exact original-coordinate conversion

$$\text{Tr}(Q_N^{\text{EW}} \dot{M}) = \text{Tr}(\Pi_N C). \quad (\text{EP9})$$

Indeed $\Pi_N C = Q_N^{\text{EW}} M M^{-1} \dot{M}$. Thus EW.31–32 and AGT4/AA23 have identical signed determinant derivatives. The positive relative operators $M(0)^{-1} M(1)$ and $M(0)^{-1/2} M(1) M(0)^{-1/2}$ are conjugate by $M(0)^{1/2}$, so EW.34 and AGT/AA use exactly the same positive relative eigenvalues and κ .

3. Existing angle proof and the exact changed pairing

The equation-level crosswalk is as follows.

Inherited EW statement	AGT statement	AA statement	Mathematical scope
EW.1–7 and EW.3–4	AGT1–3	AA1–5	Same source at EP2–4, full packet, unit, exact quotient and masses
EW.25–27 and proof of EW.28	AGT5–8	AA6, AA8–16	Same canonical projection, Gram cosines, all common directions, and signed 2×2 projection blocks
EW.31–32	AGT4	AA21–24	Same four signed derivatives after EP6–9
EW.28 rank bound and EW.35a concavity	AGT9–11, AGT15–16	AA17–26	Later crossed pairing gives $2q - 1$ instead of EW.35a's $2q$ at $n = r = q$
EW.35b–c	AGT14, AGT17–19	AA32–37	Same integration and inverse-cosh construction, with that changed count
EW.46a and its proof	AGT12–13	AA27–31	Strict positivity already proved; later direct involution proof supplies another calculation at the selected cutoffs

Inherited EW statement	AGT statement	AA statement	Mathematical scope
EW.46–48, with EW.3	AGT paragraph after AGT8	AA38–44	Same exact theta primitive; AA adds expanded division details and the explicit energy matrix coordinate

To establish the shared angle calculation without presuming a change of presentation is an identity, put $L_i = R_i E$. EW.26 proves $R_j = \Pi_j R_i$ for $i \leq j$, since both vectors have remainder x and their difference is in the relation space orthogonal to L_j . Thus

$$R_i^* M R_j = G_j, \quad G_i - G_j = (R_i - R_j)^* M (R_i - R_j). \quad (\text{EP10})$$

For isometries $F_i = R_i G_i^{-1/2}$, the two cross matrices in AGT7 and AA9 are actual adjoints:

$$F_j^* M F_i = G_j^{1/2} G_i^{-1/2}, \quad F_i^* M F_j = G_i^{-1/2} G_j^{1/2}, \quad K_{ij} = G_i^{-1/2} G_j G_i^{-1/2}. \quad (\text{EP11})$$

Their squared singular values are the eigenvalues $c_a^2 \in (0, 1]$ of K_{ij} . If $x_a \in L_i$ are orthonormal eigenvectors for the squared projection restriction and $y_a = \Pi_j x_a / c_a$, then y_a form an orthonormal basis of L_j , with $\langle x_a, y_b \rangle = c_b \delta_{ab}$. For $c_a = 1$, equality in orthogonal Pythagoras gives $x_a = y_a \in L_i \cap L_j$, and conversely every common vector contributes eigenvalue one. For $s_a = \sqrt{1 - c_a^2} > 0$, put $z_a = (y_a - c_a x_a) / s_a$. Computing these inner products gives orthonormal, mutually orthogonal two-dimensional planes, on which

$$(\Pi_i - \Pi_j)|_{\text{span}(x_a, z_a)} = \begin{pmatrix} s_a^2 & -s_a c_a \\ -s_a c_a & -s_a^2 \end{pmatrix}. \quad (\text{EP12})$$

This has eigenvalues $s_a, -s_a$; every remaining direction has eigenvalue zero. Taking determinants in EP11 yields $\log(V_i/V_j) = \sum_a -\log c_a^2$. These are precisely the calculations already written in the complete EW.26–28 proof.

EW also proves that at most $\min(q, j - i)$ of the angles move. Indeed L_i is already orthogonal to the relation columns up through degree i . The extra $j - i$ independent monic boundary columns impose at most $j - i$ linear equations on L_i , and their joint kernel is exactly $L_i \cap L_j$. Hence the common-subspace dimension is at least $q - (j - i)$ if positive. This exact rank bound includes all common vectors, and for $i = j$ all angles are zero.

At EP2, the original loss and derivative are

$$B = \log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} = \mathcal{L}_{n=q, r=q}, \quad B' = \text{Tr}((\Pi_{q-1} + \Pi_q - \Pi_{2q-1} - \Pi_{2q})C). \quad (\text{EP13})$$

EW.33–35a use the pairs $(q - 1, 2q - 1)$ and $(q, 2q)$. Each has gap q , hence its displayed estimate uses up to q moving angles and a list of length $2q$. AGT instead uses

$$(q - 1, 2q), \quad (q, 2q - 1), \quad (\text{EP14})$$

whose gaps are $q + 1$ and $q - 1$. EW's already proved rank bound gives q and $q - 1$ moving slots respectively. In AGT9 the latter is also proved directly by placing both q -dimensional spaces in the common $2q - 1$ -dimensional hyperplane. It is the same incidence bound. The reordered sum in EP13 is exactly unchanged. One zero angle from the full second list can therefore be omitted from the scalar sum, retaining every original vector space and every other zero contribution, for a count $2q - 1$.

For a self-adjoint C with width $w = c_{\max} - c_{\min}$, each positive/negative spectral pair in EP12 contributes to the trace by $s_a(\langle v_a, C v_a \rangle - \langle w_a, C w_a \rangle)$, in absolute value at most $s_a w$, since both Rayleigh quotients are in the same interval. Summing gives the shared angle estimate. For $f(x) = \sqrt{1 - e^{-x}}$, its full second derivative is

$$f''(x) = -\frac{e^{-x}}{2\sqrt{1 - e^{-x}}} - \frac{e^{-2x}}{4(1 - e^{-x})^{3/2}} = -\frac{e^{-x}(2 - e^{-x})}{4(1 - e^{-x})^{3/2}} < 0. \quad (\text{EP15})$$

The two displayed formulas in EW/AGT and AA are therefore identical. Concavity extends to zero by continuity, and finite Jensen on any retained list of a slots with total cost B gives

$$|B'| \leq a \sqrt{1 - e^{-B/a}} w. \quad (\text{EP16})$$

EW.35a gives this with $a = 2q$ at EP2, while AGT16/AA26 give it with $a = 2q - 1$. The general theorem and concavity method predate the latter specialization in the pinned source graph.

4. Full general-window consequence of the crossed pairing

The same refinement can be expressed directly in EW's general domain $n \geq q \geq 1$, $r \geq 1$, $m_{\text{EW}} = n + r$. Retain its exact $\mathcal{L} = \log(V_{n-1}V_n/(V_{n+r-1}V_{n+r}))$. Use the two nested pairs

$$(n-1, n+r), \quad (n, n+r-1), \quad a(q, r) = \min(q, r+1) + \min(q, r-1). \quad (\text{EP17})$$

For $r = 1$ the second pair is identical, contributes its full list of unit cosines, and contributes zero moving slots. In every case $a(q, r) \geq 1$. The two gap rank bounds just proved show that their nonzero angle costs fit in $a(q, r)$ slots. Append zero scalar terms if there are fewer. Their sum is the same original $\mathcal{L}$, because the two log determinant ratios telescope to the four original endpoints. The trace signs are also the same. The preceding spectral-pair estimate and EP15 therefore prove

$$|\mathcal{L}'| \leq a(q, r) \sqrt{1 - e^{-\mathcal{L}/a(q, r)}} \mathfrak{o}_H(\dot{H}). \quad (\text{EP18})$$

The count has the exact elementary form

$$a(q, r) = \begin{cases} 2 \min(q, r) - 1 = 2q - 1, & r = q, \\ 2 \min(q, r), & r \neq q. \end{cases} \quad (\text{EP19})$$

For $r < q$, the two terms are $r+1, r-1$; for $r = q$, they are $q, q-1$; and for $r > q$, both are q . This proves all cases, including $q = r = 1$. In particular the explicit improvement is available for every $n \geq q$ when $r = q$, while AGT states it at $n = q$. This paragraph is an add-only consequence derived here from EW.28 and its transport proof, and is not attributed as an extra assertion already present in the AGT body.

For any positive integer a , define the exact increasing map and its inverse by

$$\Psi_a(B) = 2 \operatorname{arcosh}(e^{B/(2a)}), \quad \Psi_a^{-1}(v) = 2a \log \cosh(v/2), \quad B, v \geq 0. \quad (\text{EP20})$$

The real branch is $\operatorname{arcosh}(z) = \log(z + \sqrt{z^2 - 1})$ for $z \geq 1$. Differentiating $\cosh(\Psi_a(B)/2) = e^{B/(2a)}$ gives $\Psi'_a(B) = [a\sqrt{1 - e^{-B/a}}]^{-1}$ for $B > 0$. For possible zero losses along a general positive C^1 family, apply the derivative to $\Psi_a(\mathcal{L} + \varepsilon)$. EP18 bounds the absolute derivative by

$$\mathfrak{o}_H(\dot{H}) \sqrt{\frac{1 - e^{-\mathcal{L}/a}}{1 - e^{-(\mathcal{L} + \varepsilon)/a}}} \leq \mathfrak{o}_H(\dot{H}). \quad (\text{EP21})$$

Integrate and let $\varepsilon \downarrow 0$ by continuity. For the original straight positive segment, put $D = (H^{(0)})^{-1}H^{(1)}$. It is self-adjoint positive for $H^{(0)}$, and every positive relative eigenvalue b gives exactly $c_b(t) = (b-1)/(1-t+tb)$. Since its derivative with respect to b is $(1-t+tb)^{-2} > 0$, the same extreme b 's determine the width throughout the path. Its integral is

$$\int_0^1 \mathfrak{o}_{H(t)}(\dot{H}) dt = \int_0^1 \frac{d}{dt} \log \frac{1-t+tb_{\max}}{1-t+tb_{\min}} dt = \log \kappa. \quad (\text{EP22})$$

Thus the crossed-pair general-window consequence is

$$|\Psi_{a(q, r)}(\mathcal{L}^{(1)}) - \Psi_{a(q, r)}(\mathcal{L}^{(0)})| \leq \log \kappa. \quad (\text{EP23})$$

It holds including zero loss for every such positive matrix pair. At $n = r = q$, it is AGT14/AA35; with $a = 2s_{\text{EW}}$, EP20 is exactly EW's Φ_s , and the proof is EW.35b's existing integration argument. This identifies both the inherited map and its changed parameter rather than giving a new name to the inherited result.

The endpoint interval is equally the same exact inverse construction. Put $A_0 = \operatorname{arcosh}(e^{\mathcal{L}^{(0)}/(2a)})$ and $d = \frac{1}{2} \log \kappa$. Equation EP23 is equivalent to

$$2a \log \cosh(\max\{0, A_0 - d\}) \leq \mathcal{L}^{(1)} \leq 2a \log \cosh(A_0 + d). \quad (\text{EP24})$$

Exponentiating and writing $z_0 = e^{A_0}$, $z_- = \max\{1, z_0/\sqrt{\kappa}\}$, $z_+ = \sqrt{\kappa} z_0$ gives

$$\left(\frac{z_- + z_-^{-1}}{2}\right)^{2a} \leq \mathcal{R}^{(1)} \leq \left(\frac{z_+ + z_+^{-1}}{2}\right)^{2a}. \quad (\text{EP25})$$

For $a = 2s$, the exponent $2a = 4s$ is exactly EW.35c. The replacement $a = 2q - 1$ is therefore a coefficient refinement of that inherited endpoint certificate. The original source-norm factor $U = \omega_{n+r}/\omega_n$ remains present in EW.20's actual endpoint $U^{1/(2r)} \sinh(\mathcal{L}/(2r))$; AGT's estimate concerns its four-volume argument and does not replace the complete EW endpoint construction.

The improvement is strict at the level of these scalar constraints for distinct positive losses. To prove it, for fixed $B > 0$ let $F_a(B) = a\sqrt{1 - e^{-B/a}}$ and set $x = B/a > 0$. Direct differentiation gives

$$\frac{\partial F_a(B)}{\partial a} = \frac{2 - (2+x)e^{-x}}{2\sqrt{1 - e^{-x}}} > 0. \quad (\text{EP26})$$

The numerator is positive because $e^x > 1 + x/2$ for $x > 0$: the difference has value zero at zero and derivative $e^x - 1/2 > 0$. Therefore $F_{2q-1}(B) < F_{2q}(B)$. Since $\Psi'_a(B) = 1/F_a(B)$, for $0 < B_0 < B_1$

$$\Psi_{2q-1}(B_1) - \Psi_{2q-1}(B_0) > \Psi_{2q}(B_1) - \Psi_{2q}(B_0). \quad (\text{EP27})$$

Integration proves this inequality; continuity gives the non-strict comparison when endpoints coincide or a zero endpoint is admitted. Thus the $2q - 1$ constraint implies the inherited $2q$ constraint and is tighter for a nonzero change of positive losses. No claim of sharpness for actual arithmetic measures or any bound uniform in growing k, q, n is inferred from this comparison.

5. Strict positivity: prior theorem and alternate later proof

EW.46a already proves $\mathcal{L} > 0$ for every admitted nonempty actual positive polynomial window. Its proof uses the actual real- u moment form. Here is its full link to the later one.

Let π_j be its real monic orthogonal polynomials, with positive squared norms ω_j . The matrix of multiplication by real u is self-adjoint. Degree and orthogonality imply

$$u\pi_j = \pi_{j+1} + b_j\pi_j + a_j\pi_{j-1}, \quad b_j \in \mathbb{R}, \quad a_j = \omega_j/\omega_{j-1} > 0 \quad (j \geq 1). \quad (\text{EP28})$$

Indeed the coefficient against π_l , $l \leq j - 2$, vanishes by moving u to π_l , whose degree is at most $j - 1$. The coefficient of π_{j-1} is $\langle \pi_{j-1}, u\pi_j \rangle / \omega_{j-1} = \omega_j/\omega_{j-1}$, since the leading coefficient of $u\pi_{j-1}$ against π_j is one. The diagonal coefficient is real by self-adjointness. The $j = 0$ equation lacks its last term.

On the original $S = c + iu$ line, the exact monic phase map $p_j(S) = i^j \pi_j((S - c)/i)$ has modulus-one factor and retains ω_j . Multiplying EP28 by its full factor gives

$$Sp_j = p_{j+1} + (c + ib_j)p_j - a_j p_{j-1}. \quad (\text{EP29})$$

If nonconstant χ divided consecutive p_j, p_{j+1} , EP29 and $a_j \neq 0$ force divisibility of p_{j-1} , and descending induction forces divisibility of $p_0 = 1$, impossible. The two residual monomial columns at degrees $N, N + 1$, after projecting away $\mathcal{P}_{N-1}$, are p_N and $p_{N+1} + zp_N$, with $z = \langle p_N, S^{N+1} \rangle / \omega_N$. Their quotient residual map F_2 cannot vanish, or both consecutive p 's would have zero remainder and be divisible by χ . Thus $F_2^* G_{N-1} F_2$ is nonzero positive semidefinite. EW.17 then gives $V_{N-1}/V_{N+1} > 1$, because the determinant of identity plus a nonzero positive semidefinite matrix is strictly greater than one. Multiplying the exact overlapping factors gives

$$\frac{V_{n-1}V_n}{V_{n+r-1}V_{n+r}} = \prod_{j=0}^{r-1} \frac{V_{n+j-1}}{V_{n+j+1}} > 1. \quad (\text{EP30})$$

This is precisely EW.46a. It applies to every positive measure along the AGT convex moment path as well, since that measure remains positive on nonzero polynomials and has every needed moment. Consequently AGT/AA's strict positivity statement already follows from the inherited proof.

AGT12–13 and AA28–30 give a different direct proof for the specific first pair in EP14. If $B = 0$, all its angles vanish, and $L_{q-1} = L_{2q}$. The first space is $\mathcal{P}_{q-1}$; the second is orthogonal to $\chi\mathcal{P}_q$. Thus $1 \perp \chi\mathcal{P}_q$. For $\chi(S) = \sum_{j=0}^q a_j S^j$, define the full antilinear involution

$$\chi^{\#k}(S) = \sum_{j=0}^q \overline{a_j}(k-S)^j. \quad (\text{EP31})$$

It has degree q , leading coefficient $(-1)^q \overline{a_q}$, and equals $\overline{\chi(S)}$ on $S = k/2 + iu$, because $\overline{S} = k - S$. Hence $1 \perp \chi \mathcal{P}_q$ would imply $0 = \langle 1, \chi \chi^{\#k} \rangle = \int |\chi(k/2 + iu)|^2 m_t(u) du > 0$. This contradiction is a shorter alternate proof at the chosen cutoffs. Its exact involution calculation is additional proof detail relative to EW's recurrence proof; the strict positivity conclusion is already inherited, in a broader window domain.

6. The theta primitive is the same original cochain

EW.46's coefficient κ_x and AA7's $X_{ij}x$ are equal under EP5. To prove this with its full coefficient metric, put $H_j = B_j^* M B_j$, using the original inclusion of B_j in the common source. Then

$$X_{ij} = H_j^{-1} B_j^* M R_i : E \rightarrow \mathcal{P}_{j-q}, \quad R_i - R_j = B_j X_{ij}, \quad G_i - G_j = X_{ij}^* H_j X_{ij}. \quad (\text{EP32})$$

The orthogonal projection onto the relation space is $B_j H_j^{-1} B_j^* M$. Applying it to R_i gives $R_i - R_j$, by EP10. This proves the second equality and then its Gram energy. For the empty relation domain the maps are its unique zero maps and no nonzero inverse is needed. Monic division makes multiplication by χ injective, so both the EW relation $R_i x - R_j x = \chi \kappa_x$ and EP32 force $\kappa_x = X_{ij} x$. In AA the name H_j denotes H_j here; EW's name H_j denotes the full source Gram from EP4. EP32 specifies the exact relationship and avoids identifying those two forms.

Retain the original $D = -x \partial_x$, $\mathcal{M} F_h = g/h$, and seed

$$\phi_0 = \phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad h(D) F_h = \Theta \phi_0. \quad (\text{EP33})$$

EW explicitly supplies the same successive divisions as AA40. Start with $F_0 = \chi(s_1 + \dots + s_k)$, and divide successively in the fixed order $s_1, \dots, s_k$: $F_{j-1} = h(s_j) L_j + F_j$, $\deg_{s_j} F_j < \deg h$. The polynomial $h(s_j)$ is monic with complex coefficients, so each division is unique over the ring in the other variables and does not increase an earlier restricted degree. The final remainder is in the tensor monomial basis of $E_h^{\otimes k}$. It has zero quotient image because χ annihilates the original sum operator on the unit; thus that remainder is zero. Therefore

$$\chi(\sum s_j) = \sum_{j=1}^k h(s_j) L_j(s). \quad (\text{EP34})$$

EW's Q_j and AA's L_j are the same coefficient polynomials if the same stated successive-division order is used. This follows inductively from the uniqueness of each monic division; every F_{j-1} is then the same polynomial in the two constructions. In particular no simple-root replacement is made and every full-order nilpotent relation remains in EP34.

For the one-leg complex $[V \xrightarrow{\Theta} \mathcal{B}]$ in degrees zero and one, both sources consequently write the identical degree- $k-1$ cochain

$$\mathcal{P}_\chi(P) = \mathcal{K}_\chi(P) = \sum_{j=1}^k (-1)^{j-1} L_j(\mathbf{D}) P(\sum D_i) (F_h^{\otimes(j-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-j)}). \quad (\text{EP35})$$

The only factor on which the tensor differential acts in the j -th summand is ϕ_0 in degree zero. There are exactly $j-1$ preceding degree-one factors, so the differential contributes $(-1)^{j-1}$, cancelling the written coefficient. Polynomial D_i 's have degree zero and commute with the differential through $D\Theta = \Theta D$. Equations EP33–34 then give, term by term,

$$d\mathcal{P}_\chi(P) = P(\sum D_i) \sum_j h(D_j) L_j(\mathbf{D}) F_h^{\otimes k} = \mathcal{V}_{h,k}(\chi P). \quad (\text{EP36})$$

This is EW.47–48 and AA42–43, with the same seed, signs, coefficients, and factor $g = 2\xi$. Substituting $P = \kappa_x = X_{ij}x$ gives precisely the same boundary of the change of canonical representative. For $k = 1$, the cyclic annihilator is the original monic h , the division gives $L_1 = 1$, and the identity reduces to EP33 with the polynomial factor $P(D)$.

EW.3 and AGT1/AA44 also preserve the exact same full-unit observation $\eta[P] = [g/h]_h^{\otimes k} P(s_1 + \dots + s_k)$. The further supported cohomology observation is the previously supplied $\sigma_h^{\otimes k} \eta$. The one-leg inclusion used in AA

maps $\phi \mapsto (\phi, 0)$ into the original two-leg complex and is the identity in degree one, so the original difference differential still sends this pair to $\Theta\phi$. It therefore carries EP35–36 to the original tau-base complex with no sign change. At every retained support label A , the map $(A, v) \mapsto (A, fv)$, $\tau \mapsto \tau$, sends a zero coefficient to the represented zero at A and external absence to external absence. This exact support rule is already stated after EW.48.

The explicitly displayed additional coordinate in AA7/AA44 is the energy $X_{ij}^* H_j X_{ij}$ in EP32 and its simultaneous presentation with the primitive. That energy is already an immediate orthogonal Pythagoras consequence of EW.11 and EW.26, as proved in EP10 and EP32. It must not be described as a new existence theorem for a theta primitive or a new connection of the angles to cohomology. Its metric is $M(t)$ along the interpolation, and the ordinary original arithmetic norm at $t = 1$; the two are related by the specified segment, not silently identified at other parameters.

7. Attribution text and cumulative integration scope

The following paragraph is suitable for R74 and the cumulative text:

The inherited endpoint-window calculation already proves the canonical principal-angle comparison, nonlinear arcosh transport, strict positivity for the actual polynomial moments, and the signed tensor theta primitive (EW.25–28, EW.35a–c, EW.46a–48). The later AGT/AA calculation independently rederives these structures and sharpens the selected window $n = r = q$: crossing the same four projection terms changes the guaranteed moving-angle count from $2q$ to $2q - 1$, yielding the corresponding stronger arcosh bound. Its direct reflected-polynomial positivity proof and explicit relation-energy coordinates provide additional proof detail. The original masses, full jets, theta primitive, and tau support remain those of the inherited construction.

If the add-only general-window consequence EP17–25 is included, append:

Applying the same crossed pairing to the inherited arbitrary-window theorem gives the count $\min(q, r + 1) + \min(q, r - 1)$. It improves the earlier count $2\min(q, r)$ by one exactly when $r = q$, for every $n \geq q$. This is an explicit refinement derived from EW’s rank bound and transport argument.

These attribution statements replace any implication that AGT or AA first supplied an all-degree principal-angle method, first obtained nonlinear arcosh transport, first proved strict moment positivity, or first constructed the tensor theta primitive. The original source has broader retained content: arbitrary windows, the norm factor U , coefficient-to-Gram maps, rational enclosures, and zero-loss positive-matrix families. That content remains part of the cumulative program; the selected-window refinement does not supersede it. The preserved global analytic limitation is the absence here of a uniform growing-packet bound on the original relative Gram or its κ . Neither source claims such a bound from the finite transport formula alone.

The companion receipt pins the compared bodies and this add-only note. The mathematical derivations above are written proofs. No numerical, Lean, remote-publication, or sealed-package mutation is claimed by this comparison.

Projection compactness in the original metric

The projection compactness calculation in the original metric coordinates. This paragraph supplies the exact coordinate map used in the compactness argument of (FS16)–(FS19). Keep the original coefficient frame of $H = \mathcal{P}_{2q}$, its actual positive Hermitian Gram $M = M(t)$ at a fixed interpolation parameter, and $n = 2q + 1$. For $0 \leq r \leq n$ put

$$\mathfrak{P}_r(M) = \{E \in \text{End}_{\mathbb{C}}(H) : E^2 = E, E^*M = ME, \text{Tr } E = r\}.$$

Here $*$ is conjugate transpose in the unchanged coefficient frame. An idempotent is the identity on its image and zero on its kernel; these spaces form a direct sum, because $x = Ex + (x - Ex)$ and their intersection is zero. Its trace therefore equals its rank. For $E \in \mathfrak{P}_r(M)$, the image and kernel are orthogonal in the original form: if $x = Ez$ and $Ey = 0$, then $x^*My = z^*E^*My = z^*MEy = 0$. Thus these are precisely the original M -orthogonal projections of rank r .

Retain the positive Hermitian square root $K = M^{1/2}$, with inverse $M^{-1/2}$. In a unitary spectral decomposition $M = V \text{diag}(\lambda_1, \dots, \lambda_n)V^*$, where all $\lambda_j > 0$, this is the concrete matrix $K = V \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})V^*$, so $K^* = K$ and $K^2 = M$. It defines the exact isometry

$$K : (H, \langle x, y \rangle_M = x^*My) \longrightarrow (\mathbb{C}^n, \langle v, w \rangle = v^*w), \quad (Kx)^*(Ky) = x^*My.$$

No source Gram or mass has been replaced: this formula retains it in the coordinate map. Conjugation and its inverse are

$$\Psi_M : \mathfrak{P}_r(M) \longrightarrow \mathfrak{P}_r(I), \quad E \longmapsto KEK^{-1}, \quad \Psi_M^{-1}(F) = K^{-1}FK.$$

To verify the first arrow in the original coordinates, the equation $E^*K^2 = K^2E$ gives

$$(KEK^{-1})^* = K^{-1}E^*K = KEK^{-1}.$$

Idempotence is preserved by conjugation, and cyclicity of the finite matrix trace gives $\text{Tr}(KEK^{-1}) = \text{Tr } E$. Conversely, if $F^* = F$, then $E = K^{-1}FK$ satisfies

$$E^*M = KFK^{-1}K^2 = KFK = ME;$$

its idempotence and trace follow by the same multiplication. The two formulas compose to the identity, and both are continuous linear maps on the finite real vector space of complex matrices. They are therefore a homeomorphism of the displayed projection sets.

The set $\mathfrak{P}_r(I)$ is closed, since it is defined by the continuous matrix equations $F^* = F$, $F^2 = F$ and $\text{Tr } F = r$. It is bounded because

$$\|F\|_{\text{HS}}^2 = \text{Tr}(F^*F) = \text{Tr}(F^2) = \text{Tr } F = r.$$

It is consequently compact in finite dimension, and it is nonempty: $\text{diag}(I_r, 0_{n-r})$ belongs to $\mathfrak{P}_r(I)$. The inverse homeomorphism proves compactness of $\mathfrak{P}_r(M)$, including $r = 0$ and $r = n$. In particular the continuous real function $E \mapsto \text{Tr}((U - W)E)$ has an attained maximum on this original projection set. Its reality follows after the same isometry from $\overline{\text{Tr}(AB)} = \text{Tr}(BA) = \text{Tr}(AB)$ for Hermitian $A = K(U - W)K^{-1}$ and $B = KEK^{-1}$. The polynomial incidence calculation in (FS14)–(FS16) excludes equality at every nontrivial rank. This compactness calculation thus justifies the strict attained inequalities of (FS17), in precisely the original metric coordinates. The notation $E^* = E$ in an orthonormal frame corresponds to $E^*M = ME$ in that original frame; the source operators, polynomial flags and spectral bounds are unchanged.

Theta-admitted analytic poles and original sources

Which analytic pole classes belong to the original theta source?

13 September 2026. This is the next calculation after AP1–26. It keeps the original one-factor arithmetic source, its Mellin measure and mass, and computes both its Hilbert admission and its strong-Schwartz admission. The resulting complexes, their residue images, their exact source maps and their theta-boundary defects are displayed. No freely chosen metric or hypothetical zero replaces the arithmetic input.

1. Original source, fixed parameters, and the exact norm

Retain the full actual packet h of degree $d > 0$, $g = 2\xi$, the entire function $v_h = g/h$, and the original F_h in the strong Schwartz source B with Mellin transform v_h . Retain the complete Taylor-unit polynomial ν representing $v_h = j_h(g/h)$; $\gcd(h, \nu) = 1$. Let P_ν be its distinct roots. At every $p \in P_\nu$ put $m_p = \text{ord}_p v_h$, a finite nonnegative integer. Since $h(p) \neq 0$, this is exactly $\text{ord}_p g$, not a multiplicity of ν or a presumed simple zero.

Use s for the Mellin coordinate and y for its critical-line imaginary part. The independent geometric parameters are $u \in \mathbb{C}^*$ and $t \in \mathbb{C}$. Distinguish

$$D_x = -x \frac{d}{dx}, \quad D_{\text{exp}} = u \frac{d}{ds} + h(s) - t, \quad L = u \frac{d}{dt} - s. \quad (\text{TA1})$$

On rational coefficients $R \in \mathbb{C}[s, \nu^{-1}]$, the stipulated original Hilbert norm is

$$\|V_{h,\text{rat}} R\|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |v_h(1/2 + iy) R(1/2 + iy)|^2 dy. \quad (\text{TA2})$$

When finite, inverse Mellin on this line defines $V_{h,\text{rat}} R$ in $L^2(\mathbb{R}_+, dx)$. Indeed logarithmic coordinates $x = e^z$ turn $F(x)$ into $e^{z/2} F(e^z)$, whose Fourier Plancherel identity is precisely TA2. No total mass is divided out. For polynomial R this is exactly $R(D_x)F_h$, the original source map.

2. Complete Hilbert admission, including zeros on the observation line

Partition P_ν into $O = \{p : \text{Re } p \neq 1/2\}$ and $C = \{p : \text{Re } p = 1/2\}$. Let $\text{PP}_p^{\leq n}$ be the span of $(s-p)^{-1}, \dots, (s-p)^{-n}$, and PP_p the union over n . The zero bound $n = 0$ means the zero space, not a discarded supported point.

The rational coefficients of finite TA2 norm are exactly

$$A^1 = \mathbb{C}[s] + \bigoplus_{p \in O} \text{PP}_p + \bigoplus_{p \in C} \text{PP}_p^{\leq m_p}. \quad (\text{TA3})$$

To prove necessity at $p = 1/2 + iy_p$, write $v_h(s) = (s-p)^{m_p} a(s)$, where $a(p) \neq 0$, and $R(s) = (s-p)^{-n} c(s)$, where $c(p) \neq 0$. Along the line, the squared modulus is bounded above and below by positive constants times $|y - y_p|^{2(m_p - n)}$ near p . Its integral is finite exactly when $2(m_p - n) > -1$, or, because the orders are integers, $n \leq m_p$. The leading coefficient cannot be cancelled by principal parts at another point.

There are no local line singularities from O . Outside a fixed compact interval, any rational R has at most polynomial growth. The actual Mellin transform v_h is rapidly decreasing on every vertical strip, as follows from F_h in B ; equivalently its gamma decay gives more than the required line decay. Thus every coefficient described in TA3 has finite norm, proving sufficiency.

The exact domain making D_{exp} an admitted differential is

$$\begin{aligned} A^0 &= D_{\text{exp}}^{-1}(A^1) \\ &= \mathbb{C}[s] + \bigoplus_{p \in O} \text{PP}_p + \bigoplus_{p \in C} \text{PP}_p^{\leq \max(m_p - 1, 0)}. \end{aligned} \quad (\text{TA4})$$

For a pole of order $n \geq 1$, $D_{\text{exp}} R$ has the uncancellable leading pole $-unc(p)(s-p)^{-n-1}$. Hence at a line point its image is admitted exactly when $n+1 \leq m_p$. A polynomial has polynomial image and is always admitted. This proves TA4 and also $A^0 \subset A^1$ without any extra domain assumption.

The two-term complex $[A^0 \xrightarrow{D_{\text{exp}}} A^1 ds]$ is consequently the original norm-admitted subcomplex of AP's analytic localization at fixed $u \neq 0$. This is a graph-domain calculation, not an assertion that all finite-norm coefficients form a differential-invariant space in both degrees.

3. Its entire cohomology and exact inclusion into the analytic family

Put $S = O \cup \{p \in C : m_p \geq 1\}$ and $Z = \{p \in C : m_p = 0\}$. Use the actual weight $w = \exp((\Phi(s) - ts)/u)$, with $\Phi' = h$ and its original constant. AP's exact residue map restricts to give

$$0 \longrightarrow H_{\text{pol}} \longrightarrow H^1(A) \xrightarrow{\text{Res}_w} \mathbb{C}^S \longrightarrow 0, \quad (\text{TA5})$$

$$0 \longrightarrow H^1(A) \longrightarrow H_{\text{loc}} \xrightarrow{\text{Res}_w|_Z} \mathbb{C}^Z \longrightarrow 0. \quad (\text{TA6})$$

Here $H^0(A) = 0$, and a basis of $H^1(A)$ is the original polynomial basis $1, s, \dots, s^{d-1}$, followed by $1/(s-p)$ for $p \in S$.

For full proof, the highest-pole argument makes D_{exp} injective on rational functions, followed by its leading-degree argument on polynomials. If an admitted F equals $D_{\text{exp}} Q$ in the larger localized complex, TA4 forces Q to belong to A^0 . Thus the induced map on H^1 has no kernel. Subtract from an admitted F its weighted residues using $(\text{Res}_w F)_p / (w(p)(s-p))$ for $p \in S$. At a point with bound m_p , AP's log-free local primitive then has pole order at most $m_p - 1$. Its sum of principal parts belongs to A^0 . Subtracting its D_{exp} image leaves a polynomial and proves the kernel assertion in TA5. Those simple-pole lifts prove surjectivity. Independence follows by residues and then polynomial cohomology. Finally AP's full residue decomposition shows that precisely the classes with zero Z -residues have such representatives, proving TA6.

All original m_p occur in the two chain spaces and in their local primitive maps. One coordinate per admitted point is the computed cohomology, not a replacement of the original zero orders by one.

Because L does not increase finite pole order and commutes with D_{exp} , it preserves A^0 and A^1 . Its connection on TA5 is exactly the submatrix of AP:

$$L = u \frac{d}{dt} - \begin{bmatrix} A(t) & B_S \\ 0 & \text{diag}(p)_{p \in S} \end{bmatrix}, \quad (\text{TA7})$$

where each column of B_S is the original constant-polynomial column. Its weighted-residue factors $w(p)$, its entire upper-right scaling integral and its oriented $2\pi i$ loop periods are the corresponding AP formulas restricted to S . No new connection or residue scaling is chosen.

4. Strong-Schwartz admission is an actual full-zero packet construction

The original source is

$$B = \left\{ F \text{ smooth on } \mathbb{R}_+ : \begin{array}{l} \sup_{x>0} x^b |D_x^j F(x)| < \infty \\ \text{for every integer } b \text{ and } j \geq 0 \end{array} \right\}. \quad (\text{TA8})$$

Its Mellin transforms are entire: for s in a compact set the bounds at zero and infinity dominate the integral and all its logarithmic derivatives. Therefore $V_{h,\text{rat}} R$ can belong to B only when every finite pole of $v_h R$ is removable, at all $p \in P_\nu$, not just on the observation line. Agreement of its entire Mellin transform with $v_h R$ on a nonsingular segment of the critical line extends to the connected punctured plane by the identity theorem, which proves the necessity of the removable-pole assertion.

Define the actual polynomials

$$\begin{aligned}
S_B &= \{p \in P_\nu : m_p > 0\}, \\
\kappa(s) &= \prod_{p \in S_B} (s-p)^{m_p}, \\
\kappa_0(s) &= \prod_{p \in S_B} (s-p)^{m_p-1}, \\
b(s) &= \kappa/\kappa_0 = \prod_{p \in S_B} (s-p), \\
c(s) &= \sum_{p \in S_B} (m_p-1)b(s)/(s-p), \\
H(s) &= h(s)\kappa(s).
\end{aligned} \tag{TA9}$$

Empty products equal one and the empty sum is zero. These are products of the specified monic factors, not renormalizations of an arbitrary leading coefficient. Since $\gcd(h, \nu) = 1$, the roots of h and κ are disjoint. H therefore selects only actual zeros of g , retaining their complete orders.

All entire-admitted rational coefficients are exactly $\kappa^{-1}\mathbb{C}[s]$. This follows either from partial fractions with the full order bounds, or by multiplying by κ : the resulting rational function has no finite poles and is polynomial. Conversely P/κ has at most the prescribed orders.

Sufficiency for the original B follows through its existing Euler inverses, without a new space-identification assumption. Apply the original S_ρ inverses successively to $\Theta\phi_*$ through every full factor of H . At each step the remaining Mellin zero supplies the required kernel condition. Their already proved source map gives $F_H \in B$ with Mellin transform g/H . Therefore

$$\begin{aligned}
V_{h,\text{rat}}(P/\kappa) &= P(D_x)F_H, \\
M(P(D_x)F_H) &= gP/H = v_h P/\kappa.
\end{aligned} \tag{TA10}$$

Every term on the right is an original strong-Schwartz theta-source vector. This proves the converse and the exact source realization, not merely finite norm. The differential graph domain of this strong-source degree-one space is $\kappa_0^{-1}\mathbb{C}[s]$, by the same highest-pole calculation as TA4 at every p .

Thus the strong-source complex is the explicit subcomplex

$$[\kappa_0^{-1}\mathbb{C}[s] \xrightarrow{D_{\text{exp}}} \kappa^{-1}\mathbb{C}[s] ds]. \tag{TA11}$$

It injects into the Hilbert-admitted and full analytic cohomologies with no extra kernel, by the same D_{exp} -preimage proof. Its residue image is exactly $\mathbb{C}^{S_B}$. In particular a ν -root off the critical line with $m_p = 0$ is admitted by the Hilbert norm but not by B ; its exact omitted coordinate is its AP residue.

5. Polynomial coordinates, source maps and every retained boundary defect

Use the literal maps $Q \mapsto Q/\kappa_0$ in degree zero and $P \mapsto P/\kappa$ in degree one. They carry TA11 to a polynomial complex with differential

$$\begin{aligned}
E(Q) &= \kappa D_{\text{exp}}(Q/\kappa_0) \\
&= ubQ' + [b(h-t) - uc]Q.
\end{aligned} \tag{TA12}$$

The quotient $\kappa'_0/\kappa_0 = \sum_p (m_p-1)/(s-p)$ proves this identity with all orders and the negative derivative term retained. Since b and h are the specified monic polynomials, the leading term of $E(Q)$ has degree $\deg Q + d + |S_B|$ and unchanged leading coefficient. Every other term has lower degree. Thus E is injective and its cokernel has the unique polynomial basis of degrees less than $d + |S_B|$, by terminating leading-term reduction.

The original polynomial exponential complex maps into it by the exact square

$$\begin{array}{ccc}
\mathbb{C}[s] & \xrightarrow{D_{\text{exp}}} & \mathbb{C}[s] \\
\text{multiplication } \kappa_0 \downarrow & & \downarrow \text{multiplication } \kappa \\
\mathbb{C}[s] & \xrightarrow{E} & \mathbb{C}[s]
\end{array} \tag{TA13}$$

$$E(\kappa_0 Q) = \kappa D_{\text{exp}} Q.$$

Leibniz differentiation proves this directly, or use the denominator maps. The original arithmetic source is carried identically, since $V_{h,\text{rat}}((\kappa P)/\kappa) = P(D_x)F_h$. No map here sends supported zero to τ or changes the base point; lift each displayed linear map on its original mask.

There is also an exact arithmetic-quotient map, with its boundary value fully retained. Write $E_H = \mathbb{C}[s]/(H)$, $v_H = j_H(g/H)$, and use the existing full packet section $\sigma_H : E_H \rightarrow Q = B/\Theta V$. For the original full jet J_H ,

$$\begin{aligned} J_H V_{h,\text{rat}}(P/\kappa) &= v_H[P]_H, \\ qV_{h,\text{rat}}(P/\kappa) &= \sigma_H(v_H[P]_H). \end{aligned} \quad (\text{TA14})$$

To justify the second equation, $H(D_x)P(D_x)F_H = \Theta(P(D_x)\phi_*)$ by Mellin identity and injectivity, so its Q -class lies in $Q[H(D_x)]$. On that actual submodule the original full-jet map and σ_H are inverse. Thus the displayed equation follows from the first one, which is the literal product of Taylor series. Restriction to the original h -block gives

$$j_h(\kappa v_H P) = v_h[P]_h. \quad (\text{TA15})$$

At each added κ -block, κP vanishes through its full order. The old source inclusion therefore retains its actual full-unit h -component and has zero added components, not a hidden mixture of old and new source classes.

For a geometric degree-zero numerator Q the observed boundary is exactly

$$qV_{h,\text{rat}}(E(Q)/\kappa) = \sigma_H(v_H[E(Q)]_H). \quad (\text{TA16})$$

It need not be zero. TA16 is the calculated defect preventing an unproved descent from geometric de Rham cohomology to the original theta quotient. The cochain source map, its full-unit jet value and its entire residual boundary have all been supplied; no false cohomology intertwiner replaces them.

At the fixed-domain specialization $u = t = 0$, $E = bh$. The quotient of the original arithmetic E_H by its ideal $(bh)/(H)$, transported by the unit v_H , is exactly the geometric polynomial quotient $\mathbb{C}[s]/(bh)$. At an added root p this ideal is $(s-p)/(s-p)^{m_p}$; at an original h -root its component is zero. This computes the complete endpoint kernel, including every removed higher jet, instead of identifying the two quotients outright. For $m_p = 1$ that component is zero. Its rank count is $d + |S_B|$, with no deletion from the larger original E_H before the specified quotient map.

The fixed pair TA11 was derived for $u \neq 0$. At $u = 0$, recomputing the maximal graph domain allows all coefficients in $\kappa^{-1}\mathbb{C}[s]$, since $h - t$ has no pole; that is a different degree-zero domain. At $u = t = 0$ its localized cohomology has rank d . The fixed-domain specialization just calculated has rank $d + |S_B|$. Both domains and the exact endpoint quotient are explicit; no flatness of the recomputed graph domain is asserted.

6. The actual theta Gram and finite source cost

For the literal denominator frame TA10, TA2 becomes the exact identity

$$\begin{aligned} \|V_{h,\text{rat}}(P/\kappa)\|^2 &= \int_{\mathbb{R}} |P(1/2 + iy)|^2 w_H(y) dy, \\ w_H(y) &= |(g/H)(1/2 + iy)|^2 / (2\pi), \quad \int w_H = \mu_H. \end{aligned} \quad (\text{TA17})$$

All removable values on the line are filled by the original entire g/H . This is the already specified arithmetic convolution seed for the actual extended packet H , with its own full mass μ_H , not a unit-mass replacement. For any finite numerator degree N , the exact Gram is

$$(M_{H,N})_{ij} = \int_{\mathbb{R}} \overline{(1/2 + iy)^i} (1/2 + iy)^j w_H(y) dy, \quad 0 \leq i, j \leq N. \quad (\text{TA18})$$

It is positive definite: a nonzero polynomial is nonzero outside a finite set, and the nonzero entire g/H is nonzero almost everywhere on the line. All moments converge by the original rapid vertical decay. In an arbitrary specified finite rational basis R_j , the unchanged norm instead has matrix $\int \overline{R_i} R_j w_h$; TA17–18 are its exact denominator-coordinate pullback, not a change of norm.

Let E_N be the coefficient matrix of TA12 from degrees $\leq N$ to degrees $\leq N + d + |S_B|$. The full source cost of the actual geometric boundary is

$$\|V_{h,\text{rat}}(E(Q)/\kappa)\|^2 = Q^* E_N^* M_{H,N+d+|S_B|} E_N Q. \quad (\text{TA19})$$

This includes u , t , every m_p through c , and all cross terms. Its arithmetic class is TA16. If G_H is an existing canonical quotient Gram at an admitted cutoff and T_N is the literal matrix $Q \mapsto v_H[E(Q)]_H$, the quotient cost is exactly $Q^* T_N^* G_H T_N Q$. No new quotient metric is selected. No uniform bound on these actual matrices is inferred from their positivity.

Completed result: all analytic pole classes have their exact original-Hilbert admission test; those with genuine strong-Schwartz source representatives are realized by an actual enlarged full-zero packet. Their complex, residues, source Gram and arithmetic boundary defect are explicitly calculated.

7. The connection in the same arithmetic numerator frame

Write $r = |S_B|$ and $q = d + r$. In the degree-one numerator frame $f_j = s^j/\kappa$, $0 \leq j < q$, the actual connection is

$$\begin{aligned} L &= u \frac{d}{dt} - A_{\text{adm}}(u, t), \\ A_{\text{adm}} &= \text{Companion}(\chi_{\text{adm}}), \\ \chi_{\text{adm}}(u, t; s) &= b(s)(h(s) - t) - uc(s). \end{aligned} \quad (\text{TA20})$$

Here Companion means the column-convention matrix taking e_j to e_{j+1} for $j < q - 1$ and taking e_{q-1} to minus the coefficient column of the lower-degree part of the displayed monic polynomial. This specifies every matrix entry without a basis convention left implicit.

Indeed $[L, E] = 0$ on parameter-dependent polynomial numerators: the term $[u d/dt, E] = -ub$ cancels $[-s, ub d/ds] = ub$, and all remaining commutators vanish. Thus L descends through the parameter-dependent cokernel. Its value on f_j is $-s^{j+1}/\kappa$. Only the last basis vector requires reduction, and the exact reduction relation is $E(1) = \chi_{\text{adm}}$. This proves TA20. It does not assert that multiplication by s alone descends on the fixed quotient by the image of the differential operator E . Nor is χ_{adm} being declared an annihilator made of new actual zeros of g . It is the calculated connection matrix in the already admitted arithmetic frame.

Here is its exact relation to AP's polynomial-plus-simple-pole frame. Use that frame restricted to S_B , and denote its connection matrix by $T_B = \begin{bmatrix} A(t) & B_{S_B} \\ 0 & \text{diag}(p) \end{bmatrix}$. Let $G(u, t)$ be the matrix whose j th column is the AP coordinates of $[s^j/\kappa]$. Its residue-block entries are

$$\begin{aligned} G_{(p,j)} &= \text{Res}_p(w(s)s^j/\kappa(s) ds)/w(p) \\ &= \frac{\left(\frac{d}{ds}\right)^{m_p-1} [w(s)s^j/\kappa_p(s)] \Big|_{s=p}}{(m_p-1)!w(p)}, \\ \kappa_p(s) &= \kappa(s)/(s-p)^{m_p}. \end{aligned} \quad (\text{TA21})$$

All m_p derivatives, the original exponential weight, and its value $w(p)$ are retained. To compute the polynomial-block entries, subtract the simple poles with these coefficients from f_j . The result has zero weighted residue at every admitted point. Take AP's log-free local primitives, keep their finite principal parts, and subtract their D_{exp} images. This leaves a polynomial. Reduce it to degree less than d with the original leading-term reduction for D_{exp} . These coefficients are exactly the remaining entries of G . This is a finite formula for each column, not a choice of an unspecified comparison map.

Both sets are proved bases of the same admitted cohomology, so G is invertible. The primitive calculations only divide by nonzero u , $w(p)$, nonzero leading coefficients, and fixed nonzero differences of distinct points. Their finite coefficients are therefore holomorphic for $u \neq 0$ and all t . Applying L to each column, with its parameter dependence included, gives the exact identity and finite scaling transport

$$\begin{aligned} uG_t &= T_B G - G A_{\text{adm}}, \\ A_{\text{adm}} &= G^{-1} T_B G - uG^{-1} G_t, \\ C_{\text{adm}}(a; t) &= G(t)^{-1} C_B(a; t) G(t + ua). \end{aligned} \quad (\text{TA22})$$

The fixed u is suppressed only in the last line's arguments. For proof of that line, differentiate its right side with respect to a , use $C'_B(a; t) = -C_B(a; t)T_B(t + ua)$, and then the first identity of TA22. The derivative is

$-C_{\text{adm}}(a; t)A_{\text{adm}}(t + ua)$ and the value at $a = 0$ is I . The uniqueness of this finite linear differential equation proves the transport formula. Both endpoint frames and the derivative gauge term are present; a pointwise similarity would omit an actual term.

These same numerator vectors are the actual source vectors $D_x^j F_H$ by TA10, so their original norm is the explicitly calculated $M_{H, q-1}$. The cohomological frame change G , however, also used D_{exp} primitives. Those primitive differences have exactly the source cost TA19 and the full arithmetic image TA16. A norm isometry between arbitrary reduced representatives is not silently inferred from a cohomology change of basis. Thus TA20–22 supply the concrete arithmetic-frame connection, its exact comparison with AP, and the already computed source-level discrepancy.

8. Exact recovery of the old canonical quotient metric

The comparison reaches the original quotient metric as well as individual source vectors. Put $K = \deg \kappa$ and $D = \deg H = d + K$. Use the literal injection

$$i_\kappa : \mathbb{C}[s]/h \longrightarrow \mathbb{C}[s]/H, \quad [P]_h \longmapsto [\kappa P]_H.$$

It is well-defined because $H = h\kappa$ and injective because the polynomial ring has no zero divisors. Its image is the ideal $(\kappa)/(H)$. At the old h -block it multiplies by the actual nonzero Taylor unit $j_h(\kappa)$; at every new block it is zero through that block's full order. Composing with v_H yields precisely the original arithmetic unit v_h by TA15, not a newly chosen identification of the h -component.

Let $R_{H,N}^{\text{coef}}$ be the existing unique least-source-norm representative for coefficient classes in $\mathbb{C}[s]/H$ among polynomials of degree at most N ; let $G_{H,N}^{\text{coef}}$ be its Gram. Define $R_{h,N-K}^{\text{coef}}$ and $G_{h,N-K}^{\text{coef}}$ in precisely the same way using the original weight w_h . Take $N \geq D - 1$ so both quotient maps are onto and both metrics are defined. Then, as maps and forms,

$$\begin{aligned} R_{H,N}^{\text{coef}} i_\kappa &= M_\kappa R_{h,N-K}^{\text{coef}}, \\ i_\kappa^* G_{H,N}^{\text{coef}} i_\kappa &= G_{h,N-K}^{\text{coef}}. \end{aligned} \tag{TA23}$$

where M_κ is literal polynomial multiplication. These are the original least-norm constructions, not newly selected quotient metrics.

For proof, a degree-at-most- N representative Q of $i_\kappa[P]_h$ has $Q - \kappa P = HT$ for some polynomial T . Thus $Q = \kappa R$ with $[R]_h = [P]_h$ and $\deg R \leq N - K$. Conversely every such R gives an allowed Q . This is a bijection between the two complete affine spaces of representatives, including all their relation directions. Their norms agree exactly:

$$\begin{aligned} \int |\kappa(s)R(s)|^2 w_H(y) dy &= \int |R(s)|^2 w_h(y) dy, \quad s = 1/2 + iy, \\ w_h(y) &= |\kappa(1/2 + iy)|^2 w_H(y). \end{aligned} \tag{TA24}$$

The last identity is $g/h = \kappa g/H$, with the removable line values filled by those entire functions. It retains both unnormalized masses; neither mass is asserted equal to the other. Finite-dimensional positive definiteness gives a unique minimizer in each affine space. The norm bijection therefore proves the first identity of TA23, and polarization proves its full Hermitian Gram identity, including cross terms.

Under the actual source maps the two minimizing polynomials give the identical strong-Schwartz function:

$$(R_{H,N}^{\text{coef}} i_\kappa[P])(D_x)F_H = (R_{h,N-K}^{\text{coef}}[P])(D_x)F_h,$$

because $\kappa(D_x)F_H = F_h$ follows from the original Mellin transforms. Thus TA23 is a calculated isometric inclusion of the original arithmetic quotient observation into the extended-packet observation with the exact degree shift K . It is stronger than a bound inferred from positivity. For precision about full-unit coordinates, write $G_{H,N}^{\text{jet}}$ for the same canonical observation expressed in physical jet coordinates $J_H = v_H \pi_H$. Then

$$\begin{aligned} G_{H,N}^{\text{coef}} &= M_{v_H}^* G_{H,N}^{\text{jet}} M_{v_H}, \\ j &= M_{v_H} i_\kappa M_{v_h}^{-1}, \\ j^* G_{H,N}^{\text{jet}} j &= G_{h,N-K}^{\text{jet}}. \end{aligned}$$

Here M_ν denotes multiplication by the indicated complete Taylor unit, not its leading coefficient. TA15 proves that j is exactly the identity on the original h -block and zero on each added full κ -block. These equations specify the equivalent metric statement in the physical jet convention used for the arithmetic cost in TA19. No unit-coordinate change is left implicit or omitted from a Gram pullback.

TA23 does not identify the extra quotient directions or their mixed Gram entries with old directions. Those remain in the actual extended metric; the geometric-boundary image TA16 and its source cost TA19 remain unchanged.

Sources and verification scope

Root reread A1–A5 of the current `tex/arithmetic_input.tex`, including the definition of B , $g = 2\xi$, $\Theta\phi_*$, and the original Euler inverse. The full A1–A19 source and its full-order section were read previously. AP1–26 is the accepted analytic input, SHA256 357923aead35c77f2fa98598fbc4ed8bc1f7cd6ebc04760da504a0ded765a652. The independent collaborator derived and checked TA3–TA7, the global graph domain and TA12–TA16 while root wrote their complete proofs and source-cost formulas here. This note claims written mathematics, not new Lean execution, numerical zero certification, a uniform estimate, or remote publication.

Complete independent theta-admission proof

Independent calculation and review of theta-admitted pole complexes

Date: 2026-09-13. This records the independent calculation supplied while the root proof `THETA_ADMITTED_POLE_COMPLEX.md` was being written. A final review seal, if appended below, identifies the complete root version read and accepted. No source-owner or publication file is edited here. There is no numerical, arithmetic-zero, Lean, or quotient-positivity certificate.

1. Exact admission for the stipulated critical-line norm

Keep the original entire function $v_h = g/h$, with h containing the full selected zero orders. Its norm on a rational coefficient F is exactly

$$\int_{\mathbb{R}} \left| v_h \left(\frac{1}{2} + iy \right) F \left(\frac{1}{2} + iy \right) \right|^2 \frac{dy}{2\pi}.$$

The parameter y on this line is distinct from the deformation parameter t in $D = u\partial_s + h - t$. Throughout the pole-complex calculation u is fixed and nonzero. No phase, cancellation, or mass in this integral is replaced by a discrete or diagonal surrogate.

Let P be the distinct roots of the actual Hermite polynomial ν , and let O and C be its off-line and critical-line subsets. At $p \in C$ put $m_p = \text{ord}_p(v_h)$. If F has an actual pole of order n at p , the weighted integrand is a nonzero smooth factor times $|y - y_p|^{2(m_p - n)}$ locally. Its integrability is equivalent to $2(m_p - n) > -1$, hence exactly to $n \leq m_p$, since both orders are integers. Apparent terms whose leading coefficients cancel must first be combined into F 's actual Laurent order.

There is no obstruction at infinity for a rational F . This can be checked from the original source without a new zeta estimate: A1–A3 give $g = \text{Mellin}(\Theta\phi_*)$ with $\Theta\phi_*$ in the strong-Schwartz space. Its Mellin restriction to any fixed vertical line is a Schwartz Fourier transform after $x = \exp(r)$. Division by h preserves this rapid decay at infinity, and the selected finite singularities are removable. Rational functions and their derivatives grow at most polynomially away from their finite poles. Off-line poles therefore impose no further condition for this line norm. Indeed every admitted weighted restriction $v_h F$ is Schwartz on the line, not merely square-integrable.

This statement proves exactly line-norm admission. It does not turn a meromorphic off-line Mellin multiplier into an entire Mellin transform.

2. The maximal two-term norm-admitted complex

Write PP_p for finite principal parts at p , and $\text{PP}_p^{\leq j} = 0$ for $j \leq 0$. The degree-one admitted rational space and its degree-zero graph domain are

$$A^1 = \mathbb{C}[s] + \bigoplus_{p \in O} \text{PP}_p + \bigoplus_{p \in C} \text{PP}_p^{\leq m_p},$$

$$A^0 = \mathbb{C}[s] + \bigoplus_{p \in O} \text{PP}_p + \bigoplus_{p \in C} \text{PP}_p^{\leq \max(m_p - 1, 0)}.$$

All sums here are direct as polynomial-plus-principal-part vector spaces. The complex is $[A^0 \xrightarrow{D} A^1 ds]$. It is the maximal such subcomplex of the analytic localization with the stipulated degree-one admission:

$$A^0 = D^{-1}(A^1) \text{ inside } \mathbb{C}[s, \nu^{-1}].$$

Indeed, at a pole of order $n \geq 1$ in Q , the highest pole of DQ has order $n + 1$ and coefficient $-un$ times the leading coefficient of Q . The multiplier $h - t$ cannot cancel it. Consequently DQ is admitted at a critical point exactly when Q has no pole or has order at most $m_p - 1$. Such a Q is automatically in A^1 as well. This proves the asserted maximal graph domain rather than assuming D preserves the original degree-one space.

Let $S = O \cup \{p \in C : m_p \geq 1\}$ and $Z = \{p \in C : m_p = 0\}$. The weighted analytic residue of AP, with $w = \exp((\Phi - ts)/u)$, has exactly the image $\mathbb{C}^S$ on the admitted complex. Its coordinates on Z vanish identically.

For every $p \in S$, the admitted lift $1/(w(p)(s-p))$ gives the p -th coordinate vector, with its actual weight retained.

If an admitted degree-one F has zero weighted residues, the AP local primitive construction gives a rational Q with $DQ - F$ polynomial. At a critical point, Q has pole order at most one less than F , so Q belongs to A^0 ; at every off-line point no additional restriction is imposed. Thus the same proof gives the exact sequence

$$0 \longrightarrow H_{\text{pol}} \longrightarrow H^1(A) \longrightarrow \mathbb{C}^S \longrightarrow 0.$$

A basis is $1, s, \dots, s^{d-1}$ followed by $1/(s-p)$ for $p \in S$. In particular $\dim H^1(A) = d + |S|$ and $H^0(A) = 0$. Full multiplicities m_p remain in the cochain spaces and their differentials; the single residue coordinate per admitted point is a proved cohomological quotient.

3. Exact map to AP and absence of an additional kernel

If an admitted F is DQ in the full analytic localization, the graph-domain identity forces Q into A^0 . Therefore the induced cohomology map is injective: restricting degree zero in this maximal way creates no extra kernel. Its image is exactly the classes whose weighted residues on Z vanish. In full, the actual map fits into

$$0 \longrightarrow H^1(A) \longrightarrow H_{\text{loc}} \longrightarrow \mathbb{C}^Z \longrightarrow 0,$$

where the last map is the corresponding coordinate restriction of AP's weighted-residue map. This also accounts for all cokernel coordinates.

There is a direct quotient-complex check. At critical p with $m_p \geq 1$, the quotient complex is

$$\text{PP}_p / \text{PP}_p^{\leq m_p - 1} \xrightarrow{D} \text{PP}_p / \text{PP}_p^{\leq m_p}.$$

Its differential is bijective: remove the highest target pole with the unique predecessor having one lower order, and continue downward; all remaining target orders at most m_p are zero in this quotient. Injectivity is the same leading-pole argument. At $m_p = 0$ the quotient is the full AP principal-part complex with its one-dimensional residue cokernel. At off-line points the quotient is zero. Hence this computation gives exactly the preceding short exact sequence and no hidden additional classes.

The operator $L = u\partial_t - s$ preserves both A^0 and A^1 and commutes with D . Multiplication by s does not raise any finite pole order, and t differentiation does not alter these fixed pole bounds. The admitted connection, weighted-residue transition, and scaling matrix are therefore AP's matrices with precisely the pole rows and columns in S retained: $\begin{bmatrix} A(t) & B_S \\ 0 & P_S \end{bmatrix}$, W_S , and their displayed off-diagonal integral. This is an invariant subfamily, not a chosen replacement operator.

4. Entire-Mellin admission and the actual strong-Schwartz source

For this stronger admission put $m_p = \text{ord}_p(v_h)$ at every p in P , not just on the critical line. Since $\gcd(h, \nu) = 1$, h does not vanish at these points and this equals $\text{ord}_p(g)$. Let

$$\begin{aligned} k &= \prod_{m_p > 0} (s-p)^{m_p}, \\ k_0 &= \prod_{m_p > 0} (s-p)^{m_p - 1}, \\ b = k/k_0 &= \prod_{m_p > 0} (s-p). \end{aligned}$$

The empty products are one. The rational coefficients F for which $v_h F$ is entire are exactly $k^{-1}\mathbb{C}[s]$. Their graph domain under D for u nonzero is $k_0^{-1}\mathbb{C}[s]$, by the same leading-pole argument. The inclusion into the line-norm admitted complex is literal. Its image on cohomology retains just the residues at points with $m_p > 0$, and it is injective by the same maximal-domain argument. Thus the difference from the line-norm domain consists exactly of off-line points where $m_p = 0$.

This stronger space has an exact realization in the original source, not only a formal entire-function criterion. Set $H = hk$. Every root of H is an actual g -zero with its full order, and its two factors have disjoint root sets.

The original repeated-root division construction gives $F_H = S_H \Theta \phi_*$ in the strong-Schwartz space, with Mellin transform g/H . Consequently

$$V(P/k) = P(D_x)F_H, \text{ where } D_x = -x\partial_x,$$

has Mellin transform $gP/H = v_h P/k$. This proves sufficiency by the original source construction. Necessity follows because the Mellin transform of a strong-Schwartz source is entire: every pole of $v_h F$ must cancel. The coefficient derivative ∂_s in D is not D_x ; the latter corresponds to multiplication by s under Mellin transform.

The full source jet map is

$$J_H V(P/k) = v_H[P]_H,$$

with the actual complete unit $v_H = j_H(g/H)$. On the old polynomial inclusion the target numerator is kP ; restricting to the original h packet gives $j_h(kv_H P) = v_h[P]$. At newly added roots kP vanishes to the full corresponding orders. These identities retain all original jets and the full Taylor units; they do not replace them by leading values.

5. Polynomial coordinates for the stronger complex

Under the degree-zero and degree-one denominator identifications, put

$$c = bk'_0/k_0 = \sum_{m_p > 0} (m_p - 1)b/(s - p).$$

This is a polynomial, including when some or all m_p equal one. The exact polynomial differential is

$$E = ub\partial_s + b(h - t) - uc,$$

because $kD(P/k_0) = E(P)$. The old polynomial complex embeds by multiplication by k_0 in degree zero and k in degree one:

$$E(k_0 P) = kD(P).$$

If $q = d + \deg b$, the leading term of $E(P)$ for nonzero polynomial P has degree $\deg P + q$ and unchanged leading coefficient. The derivative and c terms have strictly smaller degree. Hence E is injective, its cohomology is free on the degree-below- q remainder basis at fixed parameters, and its dimension q agrees with the exact residue sequence.

The commuting connection in this numerator basis is also explicit. $[u\partial_t - s, E] = 0$, and the last basis column is reduced using $E(1) = b(h - t) - uc$. Thus its matrix is

$$u\partial_t - \text{Companion}(b(h - t) - uc).$$

It need not be pointwise similar to AP's polynomial-plus-simple-poles block: their parameter-dependent frame transition is a connection gauge, whose derivative term must be included. In particular c records the full multiplicities without changing the original arithmetic packet H .

These coefficient/source maps are degreewise. An exponential boundary $E(P)$ need not be an arithmetic theta boundary; its exact arithmetic jet observation is $v_H[E(P)]_H$. No descent through the arithmetic quotient is asserted by merely writing a strong-Schwartz representative. Likewise the fixed pair with denominators k_0, k is the maximal graph domain only when u is nonzero. Setting $u = 0$ in its matrix is not the same as recomputing the maximal graph domain, which then admits the full orders m_p in degree zero as well.

Source and verification scope

The original source identities A1–A3 were reread in `arithmetic_input.tex`. Its previously recorded SHA-256 is `a50b0fb587b644c4ea94dba45b2b423d038f2bfbef7fef188543939ba850e58d2`. The repeated-root primitive, full-unit section, and exact original source were read completely in the prior independent scaling work. The accepted AP proof supplies the already independently checked local weighted-residue primitive and entire transport, with no change of arithmetic input.

All cohomology assertions above are algebraic. Finite norm of the displayed representatives does not by itself prove closedness of the boundary range or a nondegenerate induced quotient norm. This review neither chooses a new metric nor asserts such a conclusion.

Full read of the root TA1–19 proof

The complete root TA1–19 note was read after it became available, including its source, endpoint, and cost paragraphs. No mathematical correction was required in that version. In addition to the independent calculations above, the following root extensions were checked directly.

First, the original source identification descends exactly to its stated arithmetic-quotient value. Multiplying $P(D_x)F_H$ by $H(D_x)$ gives $\Theta(P(D_x)\phi_*)$, so its Q -class belongs to the actual H -torsion submodule. The original packet section is inverse to the full jet map there. This proves the root's $qV(P/\kappa) = \sigma_H(v_H[P]_H)$ with the complete unit. Applying the same identity to $E(Q)$ proves the exact nonzero boundary observation; no assertion of geometric-boundary descent is needed or made.

Second, at the fixed-pair endpoint $u = t = 0$ the geometric polynomial differential is multiplication by bh . Since bh divides H , the map $\mathbb{C}[s]/H \rightarrow \mathbb{C}[s]/(bh)$ is onto with kernel $(bh)/(H)$. At an old h -root its kernel is zero, and at an added p it is exactly $(s - p)/(s - p)^{m_p}$. The kernel dimension is $\sum_{p \in S_B} (m_p - 1)$, and the quotient dimension is $d + |S_B|$. Multiplication by the actual unit preserves this ideal and gives the stated transported comparison. Recomputing the graph domain at $u = t = 0$ instead uses κ in both degrees and produces the rank- d quotient. This verifies both endpoint statements with their distinct domains.

Third, TA17–19 are exact pullbacks of the retained source norm. For numerator P the weight is $|g/H|^2/(2\pi)$, with the original unrenormalized mass. Its Gram is positive on every finite polynomial coefficient space because g/H is a nonzero entire function and a nonzero polynomial has only finitely many zeros. All moments converge by the original source's rapid vertical decay. For a fixed matrix E_N , expanding the original quadratic form gives precisely $Q^*E_N^*M_H E_N Q$, including every cross term. Composing instead with the existing arithmetic quotient map and its specified G_H gives $Q^*T_N^*G_H T_N Q$. These equalities are changes of coordinates in given forms; they do not define a new quotient norm or infer a uniform estimate from positive definiteness.

Complete review of TA20–22 and the frame transport

The full new Section 7 was read, including the explicitly stated column companion convention, the finite primitive construction of G , its holomorphic parameter domain, and the source-discrepancy paragraph. Its formulas agree exactly with the independent connection calculation.

For each numerator vector s^j/κ , the simple-pole AP coefficient at p is its weighted residue divided by $w(p)$. Expanding κ at its full order m_p gives exactly TA21's derivative of order $m_p - 1$ and factorial $(m_p - 1)!$. The complementary factor κ_p is nonzero at p . After subtracting those simple poles, AP's primitive reduces the remaining zero-residue rational coefficient within the admitted degree-zero bound. Polynomial leading-term reduction then supplies its unique old-polynomial coordinates. This proves that the displayed finite algorithm computes G with every required unit, derivative, and residue value retained.

The two bases have already been proved independent and spanning, hence G is invertible without a separate determinant assumption. The finite algorithm has only the stated nonzero denominators on u nonzero and all t ; a value $h(p) - t = 0$ does not introduce an unlisted division. Thus the asserted holomorphic parameter domain is correct, including the empty-pole case in which G is the identity.

The frame orientation is AP coordinates = G times numerator coordinates. Applying L gives $uG_t - T_B G = -GA_{\text{adm}}$, which is TA22 with exactly the negative derivative term in $G^{-1}T_B G - uG^{-1}G_t$. Differentiating $G(t)^{-1}C_B(a; t)G(t + ua)$ therefore yields $-C_{\text{adm}}(a; t)A_{\text{adm}}(u, t + ua)$, with initial value identity. This checks the matrix order, signs, source and target parameter fibres, and both endpoint frame factors. No pointwise similarity is substituted for the connection gauge.

The representative differences used to construct this cohomological frame change are actual D_{exp} primitives in the strong-source graph domain. Their source images consequently have precisely the already computed TA19 cost and TA16 arithmetic boundary value. Keeping this discrepancy prevents an unjustified source-norm isometry or arithmetic cohomology descent from being inferred from G .

Final independent acceptance

The complete root proof TA1–22, including all unnumbered scope statements and the final source paragraph, has now been read and checked. The exact accepted artifact is THETA_ADMITTED_POLE_COMPLEX.md, SHA-256:

3f9a79e61de1f38bc0944be4e43b1ef20f02f99b55637f3afa8b773fb38cf8ec.

Final disposition: accept TA1–22 with no outstanding mathematical correction. This is a written-proof review of the norm and strong-source admission, both maximal graph complexes, exact residue images and injections, full-packet source realization, retained arithmetic boundary defect, endpoint quotient, actual Gram/cost pullbacks, and connection frame transport. No numerical fixture, arithmetic-zero evaluation, Lean replay, uniform metric bound, or new Frobenius identification is certified by this acceptance.

Additive review of the exact canonical-metric recovery, TA23–24

The complete Section 8 was subsequently read, including its explicit coefficient-versus-physical-jet conventions. The preceding TA1–22 seal and its hash remain a historical acceptance of that earlier version. The following checks concern the additive TA23–24 continuation.

Set $K = \deg \kappa$, so K includes every full added zero order, not merely the number of distinct added points. With $D = \deg H = d + K$, the stated cutoff $N \geq D - 1$ gives $N - K \geq d - 1$. Both coefficient quotient observations are therefore onto, including the endpoint cutoffs and the empty-added- packet case $K = 0$. Multiplication by κ is injective from $\mathbb{C}[s]/h$ to $\mathbb{C}[s]/H$ because $H = h\kappa$ in the polynomial integral domain.

Fix a coefficient class $[P]_h$. A representative Q of its image satisfies $Q - \kappa P = HT$ if and only if $Q = \kappa R$ with $[R]_h = [P]_h$. Because multiplication by the nonzero degree- K polynomial κ shifts the degree of every nonzero R by exactly K , the restriction $\deg Q \leq N$ is exactly $\deg R \leq N - K$. Zero is included separately with no difficulty. This proves a bijection of the whole affine representative spaces, not only an injection of selected representatives or a comparison inequality.

The pointwise identity $v_H \kappa = v_h$ gives equality of the original norms for each pair of corresponding representatives, with their original measure and neither source mass renormalized. Positive definiteness of the finite coefficient Grams gives unique minimizers; the bijection therefore proves the representative identity in TA23. Polarization proves its full Hermitian-form identity, retaining all mixed terms among old quotient directions. The actual source vectors coincide because $\kappa(D_x)F_H = F_h$ by Mellin injectivity. Thus there is no unrecorded source representative or cutoff defect.

The addendum correctly distinguishes G^{coef} from the physical-jet G^{jet} used in TA19. For $U_H = M_{v_H}$ and $U_h = M_{v_h}$, the identities

$$\begin{aligned} G_H^{\text{coef}} &= U_H^* G_H^{\text{jet}} U_H, \\ j &= U_H i_\kappa U_h^{-1} \end{aligned}$$

give $j^* G_H^{\text{jet}} j = G_h^{\text{jet}}$ at the stated respective cutoffs. On the original h -block, $v_H \kappa = v_h$ proves that j is identity. On every added block it is zero through the full order because κ vanishes to that order and both units are genuine full Taylor units. This is precisely the original packet's extension by zero in physical jets, not the bare coefficient multiplication map silently reused in a different coordinate convention.

No orthogonality to the added directions follows or is claimed. Their mutual and mixed Gram entries remain in the extended metric. TA16's arithmetic boundary value and TA19's full cost are unchanged. Therefore the addendum establishes an exact restriction of the existing canonical metrics, without selecting a new metric or deriving a uniform estimate.

Final acceptance of TA1–24

The complete added Section 8 and its final unit-coordinate clarification have been read and checked. The exact accepted proof is THETA_ADMITTED_POLE_COMPLEX.md, SHA-256:

830e9a254465434fd61cb16212544f004ddf54a5adc72554bccabe217b045e5a.

Final disposition: accept TA1–24 and their stated scope, with no outstanding mathematical correction. This supersedes only the version pin of the earlier TA1–22 acceptance, which remains recorded above. The present addition is a written-proof review of exact cutoff, minimizer, source-map, and full-unit Gram identities; it includes no numerical replay, Lean execution, arithmetic-zero evaluation, or uniform metric-bound claim.

Added arithmetic directions and exact mixed volume

The added arithmetic directions: exact mixed metric and volume

13 September 2026. This continues TA1–24 on the original theta source. TA23 recovers the old packet's canonical metric inside an extended actual packet. Here the other directions, their mixed terms and their quotient volume are calculated. The quotient behind the Schur complement is identified by its original source map; no new metric is selected.

1. Actual packets, original sources, and fixed coordinate conventions

Use TA's coprime monic actual full-zero packets h and κ , $H = h\kappa$, with $d = \deg h > 0$, $K = \deg \kappa$, and $D = d + K$. The multiplicities are their full orders as zeros of $g = 2\xi$. This does not assert a new zero exists. $K = 0$ is allowed and treated at the end. Initially take $K > 0$.

For $f = h, \kappa, H$, retain the original Euler-inverse source F_f with Mellin transform g/f , original generator $D_x = -x\partial_x$, and density

$$w_f(y) = |(g/f)(1/2 + iy)|^2 / (2\pi).$$

The original strong-Schwartz source gives all finite moments and positive finite polynomial Grams. None of the masses $\int w_f$ is divided out. Write $P_n = \mathbb{C}[s]_{\leq n}$, $E_f = \mathbb{C}[s]/f$, in increasing-power remainder bases. For $N \geq D - 1$, all quotient observations below are onto. Let $R_{f,n}^{\text{coef}}$ and $G_{f,n}^{\text{coef}}$ denote the original unique least-source-norm representative and its Gram for E_f from P_n with weight w_f . Abbreviate

$$A = G_{h,N-K}^{\text{coef}}, \quad B = G_{\kappa,N-d}^{\text{coef}}, \quad G = G_{H,N}^{\text{coef}}.$$

The superscript coef refers to coefficient remainders, not physical jets. Physical jets retain the full unit $U_f = M_{v_f}$, $v_f = j_f(g/f)$, and $G_{f,n}^{\text{coef}} = U_f^* G_{f,n}^{\text{jet}} U_f$.

2. Full block matrix and the exact mixed term

Define the literal polynomial map

$$\begin{aligned} T : E_h \oplus E_\kappa &\longrightarrow E_H, \\ T(P, Q) &= \kappa P + hQ. \end{aligned} \tag{EM1}$$

It is well-defined. Reduction modulo h and modulo κ proves it is bijective, since $\kappa \bmod h$ and $h \bmod \kappa$ are units. Every original jet order is retained by these reductions. TA23 applied to both factors gives the exact representatives and block Gram

$$\begin{aligned} R_{H,N}^{\text{coef}} T(P, Q) &= \kappa R_{h,N-K}^{\text{coef}} P + h R_{\kappa,N-d}^{\text{coef}} Q, \\ T^* G T &= \begin{bmatrix} A & C \\ C^* & B \end{bmatrix}. \end{aligned} \tag{EM2}$$

For completeness, each summand on the right is the canonical lift of its own class by TA23. Their sum is still orthogonal to the full relation space HP_{N-D} . It is therefore the unique canonical lift of the sum. This proves the first identity, including its representative, not only its quotient value. Taking all pairwise original source inner products proves the second. Specifically, in the indicated bases,

$$\begin{aligned} C_{ij} &= \int \overline{\kappa(s)(R_{h,N-K}^{\text{coef}} e_i)(s)} h(s)(R_{\kappa,N-d}^{\text{coef}} e_j)(s) w_H(y) dy, \\ s &= 1/2 + iy. \end{aligned} \tag{EM3}$$

Equivalently C_{ij} is the original source inner product between $(R_{h,N-K}^{\text{coef}} e_i)(D_x)F_h$ and $(R_{\kappa,N-d}^{\text{coef}} e_j)(D_x)F_\kappa$. This follows from the exact identities $\kappa(D_x)F_H = F_h$ and $h(D_x)F_H = F_\kappa$. Every phase and mixed term in EM3 is retained. No orthogonality of distinct zero blocks is assumed.

3. The original source quotient behind the Schur complement

Let $U = M_{h \bmod \kappa}$ on E_κ . Define the canonical Gram Γ_N directly from the following observation of the same source P_N, w_H :

$$0 \longrightarrow P_{N-K} \xrightarrow{\text{multiplication } \kappa} P_N \xrightarrow{\pi_\kappa} E_\kappa \longrightarrow 0. \quad (\text{EM4})$$

This is an explicit additional observation of the already fixed source, not a replacement of its norm by w_κ . Its unique least-norm section will be denoted R_{free} . Then the exact Schur complement and minimizer are

$$\begin{aligned} S_N &= B - C^* A^{-1} C = U^* \Gamma_N U, \\ P_{\min} &= -A^{-1} C Q, \\ \kappa R_{h, N-K}^{\text{coef}} P_{\min} + h R_{\kappa, N-d}^{\text{coef}} Q &= R_{\text{free}}(UQ). \end{aligned} \quad (\text{EM5})$$

Here is a proof through the complete original affine spaces. First, $\pi_\kappa T(P, Q) = hQ = UQ$. For fixed Q , varying P in E_h ranges through every class of E_H with this prescribed κ remainder: the kernel of $E_H \rightarrow E_\kappa$ is precisely the ideal $(\kappa)/(H)$, the image of the first summand of T . Each of these classes in turn has all its P_N representatives obtained by adding HP_{N-D} . Their union is exactly every polynomial in P_N with κ remainder UQ . Thus minimizing first within each H -class and then over P is exactly the minimization specified by EM4, with no omitted relation directions.

In EM2 the squared norm at fixed Q is

$$(P + A^{-1} C Q)^* A (P + A^{-1} C Q) + Q^* (B - C^* A^{-1} C) Q.$$

Since A is positive definite, the minimum has the stated unique $P_{\min}$. This proves the form identity and, by uniqueness, the source identity in EM5. Γ_N and S_N are positive definite; the larger relation space in EM4 has not been replaced by the smaller H -relation space.

In the original arithmetic cohomology $Q_\theta = B_{\text{source}}/\Theta V$, the full packet has its original section σ_H and full-jet observation. Projection onto the κ -primary arithmetic submodule is the actual map

$$\begin{aligned} Q_\theta[H(D_x)] &\longrightarrow Q_\theta[\kappa(D_x)], \\ z &\longmapsto \sigma_\kappa \text{pr}_\kappa J_H z, \\ P(D_x)F_H &\longmapsto \sigma_\kappa(v_H|_\kappa[P]_\kappa). \end{aligned} \quad (\text{EM6})$$

The last expression means the quotient class of that source vector. The previous equality follows from the full-jet inverse on the actual H -torsion submodule. Its kernel is the original h -primary submodule. Consequently EM4–5 calculate the finite source cost of the same arithmetic projection, expressed in its stated coordinates. These are the arithmetic submodules of the original τ -base pushforward; no different cohomology group or source sheaf is substituted.

4. Resultant, determinant line, and the full-unit conversion

In the increasing-power bases, T 's columns are

$$\kappa, \kappa s, \dots, \kappa s^{d-1}, h, hs, \dots, hs^{K-1}.$$

Its determinant is exactly $\text{Res}(h, \kappa)$, including sign in this convention. Indeed the basis $[1, s, \dots, s^{d-1}, h, hs, \dots, hs^{K-1}]$ has triangular change-of-basis determinant one. In that basis T has diagonal blocks $M_{\kappa \bmod h}$ and I_K . The determinant of the former is $\prod_{h(\rho)=0} \kappa(\rho)^{\text{ord}_\rho h} = \text{Res}(h, \kappa)$, with complete multiplicities. It is nonzero by coprimality. Hence EM2 gives

$$\det G = \det A \det S_N / |\text{Res}(h, \kappa)|^2. \quad (\text{EM7})$$

This is the determinant-line factor for the displayed ordered map, not an arbitrary volume convention. Also $\det U = \text{Res}(\kappa, h)$, and $\text{Res}(\kappa, h) = (-1)^{dK} \text{Res}(h, \kappa)$. Substituting the full U term of EM5 therefore yields the further exact identity

$$\det G_{H,N}^{\text{coef}} = \det G_{h,N-K}^{\text{coef}} \det \Gamma_N. \quad (\text{EM8})$$

Both resultant factors were retained and cancel algebraically in EM8. Neither is individually set to one.

For the physical full-jet formulation, put $a = v_H|_\kappa$ and

$$\begin{aligned} J &= U_H T \text{diag}(U_h^{-1}, U_\kappa^{-1}), \\ \Gamma_N^{\text{phys}} &= M_a^{-*} \Gamma_N M_a^{-1}. \end{aligned} \quad (\text{EM9})$$

The columns of J are exactly the two extension-by-zero inclusions of the full physical h and κ jets, by $v_h = \kappa v_H|_h$ and $v_\kappa = h v_H|_\kappa$. They are not asserted orthogonal. The physical Schur complement is Γ_N^{phys} , since $U_\kappa = M_a U$ and the multipliers commute. The absolute determinant of J is $1/|\text{Res}(h, \kappa)|$: the determinant of U_H is the product of its two restriction determinants, whereas $\det U_h$ times $\det U_\kappa$ has the two additional resultant factors. Thus the physical formula is

$$\det G_{H,N}^{\text{jet}} = |\text{Res}(h, \kappa)|^2 \det G_{h,N-K}^{\text{jet}} \det \Gamma_N^{\text{phys}}. \quad (\text{EM10})$$

This preserves the exact coordinate change from the coefficient formula. Physical jets in monomial remainder coordinates are not assigned a product determinant frame without this J factor.

5. The relation-volume is the original theta seed

Define, for $f = h, H, 1$, the actual moment Gram determinants

$$\begin{aligned} \Delta_{f,n} &= \det \left[\int \overline{s^i} s^j w_f(y) dy \right]_{i,j=0}^{n-1}, \\ \Delta_{f,0} &= 1, \quad w_1 = |g|^2/(2\pi), \quad s = 1/2 + iy. \end{aligned}$$

The original relation source for E_f at cutoff $n-1$ consists of f times polynomials. Its weight is exactly $|f|^2 w_f = w_1$. In particular the relation weight does not depend on which actual full-zero packet was divided out. Using the monic ordered basis of quotient powers followed by f times relation powers, whose determinant is one, the ordinary Gram Schur complement proves, for $n \geq \deg f$,

$$\det G_{f,n-1}^{\text{coef}} = \Delta_{f,n} / \Delta_{1,n-\deg f}.$$

Apply this at the actual paired cutoffs in TA23. The two relation dimensions are both $N - D + 1$, so their full denominators agree, giving

$$\det G_{H,N}^{\text{coef}} / \det G_{h,N-K}^{\text{coef}} = \Delta_{H,N+1} / \Delta_{h,N-K+1} = \det \Gamma_N. \quad (\text{EM11})$$

The last equality also follows directly from EM4: its relation Gram has weight $|\kappa|^2 w_H = w_h$ and dimension $N - K + 1$. Thus EM11 is verified by the two exact sequences independently. At $N = D - 1$ the common original H/h relation dimension is zero and its determinant is one; no negative index is introduced. All numerator and denominator masses remain those of the displayed arithmetic weights.

This identifies the added quotient volume by two explicitly specified arithmetic moment determinants. It is not left as the determinant of an unnamed complement.

6. Exact mixed-volume cost

One can also retain the entire loss relative to keeping the added coordinates fixed with the old coordinates zero:

$$\begin{aligned} \det S_N / \det B &= \det(I_K - B^{-1} C^* A^{-1} C), \\ Q^*(B - S_N)Q &= (A^{-1} C Q)^* A (A^{-1} C Q). \end{aligned} \quad (\text{EM12})$$

The endomorphism $B^{-1} C^* A^{-1} C$ is self-adjoint and nonnegative in the existing B form; its eigenvalues belong to $[0, 1)$, because EM5 is positive definite. These assertions follow respectively by taking its B inner product and by

writing $Q^*S_NQ > 0$. The right-hand side of the second identity is the exact original source cost of the optimizing old component. No change to a unit-mass measure or a different metric has been made. The determinant loss is the product of the retained 1-minus-eigenvalue factors, not a presumed zero mixed term.

These identities compute the finite extension volume and all mixed contributions. They do not prove a uniform bound as packet size and cutoff vary. The original full arithmetic-boundary discrepancy TA16, its source cost TA19 and the geometric connection TA20–22 remain in place; projecting physical arithmetic classes in EM6 is not an assertion that every geometric exponential boundary has become a theta boundary.

For $K = 0$, E_κ is zero, all its determinants are one, the second block and its maps are empty, $\text{Res}(h, 1) = 1$, and $H = h$. EM1–12 reduce to the identity comparison with no added classes. No nonempty extension is inferred from the formulas.

Sources and verification scope

The actual-source input is the accepted TA1–24 proof, SHA256 830e9a254465434fd61cb16212544f004ddf54a5adc72554bccabe217b045e5a. Root reread its final metric section and the original arithmetic-input Euler inverse, full jets and packet section A5–A10. Root also reread AT1–AT7: the original τ pushforward, retained source map and canonical Gram construction. These are applied here to the specified one-factor arithmetic packets. No tensor-sum convolution identity is inferred by replacing its separate cyclic relation with H . This note contains written proofs and no new Lean execution, interval computation, or zero-location certificate. An independent collaborator derives and checks the block and quotient maps separately; its final review pins the accepted text.

Complete independent metric-extension proof

Independent review of the canonical metric extension

Date: 2026-09-13. This records the independent calculation supplied while the root proof `MAIN_METRIC_EXTENSION.md` was being written. A final seal, if appended, identifies the complete root artifact read and accepted. The accepted input is TA1–24 with $H = h\kappa$, both factors retaining their full actual zero orders, and $\gcd(h, \kappa) = 1$. This is local written mathematics only; no new zero, metric, normalization, numerical certificate, Lean execution, or remote change is asserted.

1. Source spaces, cutoffs, and notation

Write $d = \deg h$, $K = \deg \kappa$, $D = d + K$, and take $N \geq D - 1$. The source coefficient space is $P_N = \mathbb{C}[s]_{\leq N}$ (with $P_n = 0$ for $n < 0$) and its original positive form

$$\langle F, G \rangle_H = \int_{\mathbb{R}} \overline{F(1/2 + iy)} G(1/2 + iy) w_H(y) dy,$$

where $w_H = |g/H|^2/(2\pi)$. Let π_H be remainder modulo H . Its canonical least-source-norm section is $R_{H,N}$ and its coefficient-quotient metric is $G_{H,N}$. Every G without a superscript in the calculation below is in coefficient coordinates. The source function of F is $F(D_x)F_H$.

Similarly retain $R_{h,N-K}$ and $G_{h,N-K}$ for the original weight $w_h = |\kappa|^2 w_H$, and $R_{\kappa,N-d}$ and $G_{\kappa,N-d}$ for $w_\kappa = |h|^2 w_H$. No weight is normalized to total mass one. Both quotient cutoffs are sufficient because $N - K \geq d - 1$ and $N - d \geq K - 1$.

The monic polynomials give the literal isomorphism

$$\begin{aligned} T : E_h \oplus E_\kappa &\longrightarrow E_H, \\ T(P, Q) &= \kappa P + hQ. \end{aligned}$$

It is well defined. If its value vanishes modulo $h\kappa$, reduction modulo h and then κ , using coprimeness, forces $P = Q = 0$ in the respective quotient algebras. Equal dimensions then prove surjectivity.

2. The complete mixed block is an original source pairing

TA23 applied to both factors gives

$$R_{H,N}T(P, Q) = \kappa R_{h,N-K}P + h R_{\kappa,N-d}Q.$$

This uses linearity of the canonical section as well as both exact restriction identities. Thus, suppressing the now fixed cutoffs,

$$T^*G_H T = \begin{bmatrix} G_h & C \\ C^* & G_\kappa \end{bmatrix},$$

where, in any specified coefficient bases,

$$C_{ij} = \int \overline{\kappa R_h e_i} (h R_\kappa e_j) w_H,$$

or equivalently

$$C_{ij} = \langle (R_h e_i)(D_x)F_h, (R_\kappa e_j)(D_x)F_\kappa \rangle_{L^2(dx)}.$$

Both source identities follow from the original Mellin transforms. The integral retains the complex phase of $\bar{\kappa}h$, every source mass, and all the cross terms. Neither orthogonality of the two packet blocks nor equality of their total masses is assumed.

3. Resultant and Schur determinant with full multiplicities

In ascending monomial bases, the columns of T are $\kappa, s\kappa, \dots, s^{d-1}\kappa$, followed by $h, sh, \dots, s^{K-1}h$. This is the coefficient Sylvester map, with

$$\det T = \text{Res}(h, \kappa)$$

for the convention $\text{Res}(h, \kappa) = \prod_{h(\alpha)=0} \kappa(\alpha)$ with multiplicity and monic h . To check the convention explicitly when both root sets are distinct, evaluate the columns at the h -roots followed by the κ -roots. The two diagonal evaluation blocks have determinants $\text{Res}(h, \kappa) \det V_h$ and $\text{Res}(\kappa, h) \det V_\kappa$; the full evaluation matrix has determinant

$$(-1)^{dK} \text{Res}(h, \kappa) \det V_h \det V_\kappa.$$

Using $\text{Res}(\kappa, h) = (-1)^{dK} \text{Res}(h, \kappa)$ gives the stated sign. Both sides are polynomials in the original coefficients, so the identity holds with repeated roots as well. In particular coprimeness makes it nonzero, without removing any multiplicity.

Let

$$S = G_\kappa - C^* G_h^{-1} C.$$

Completing the square in the full positive form proves that S is positive definite and

$$\det G_H = \frac{\det G_h \det S}{|\text{Res}(h, \kappa)|^2}.$$

The squared absolute determinant here is essential: the Gram pullback is Hermitian, not a bilinear determinant transformation.

4. The actual quotient measured by the Schur complement

Define Γ as the canonical quotient metric of the SAME source P_N with weight w_H under the map $\pi_\kappa : P_N \rightarrow E_\kappa$. Its relation space is exactly

$$\ker \pi_\kappa = \kappa P_{N-K}.$$

This is an existing least-source-norm construction with the actual source and a specified observation, not an arbitrary positive matrix. It has cutoff N , not $N - d$, and source weight w_H , not w_κ . Write its canonical section as $R_{\kappa, N}^{w_H}$. Put

$$U = M_{h \bmod \kappa} : E_\kappa \rightarrow E_\kappa.$$

The exact projection identity is

$$\pi_\kappa T(P, Q) = UQ.$$

Consequently, for fixed Q , allowing P to vary is exactly allowing all old h -jet coordinates to vary while fixing the κ remainder to UQ . Taking the source infimum first over full H representatives and then over P is the same as taking it over the single complete source fibre $\{F \in P_N : \pi_\kappa F = UQ\}$. This proves

$$S = U^* \Gamma U.$$

It is not correct to identify S with bare Γ in T 's Q coordinates; the h -unit U is part of the exact quotient map. There is also an explicit minimizing source map. The unique minimizing old coefficient is

$$P_* = -G_h^{-1} CQ.$$

The corresponding polynomial and source function are

$$R_{\kappa, N}^{w_H}(UQ) = \kappa R_h P_* + h R_\kappa Q,$$

$$(R_h P_*)(D_x)F_h + (R_\kappa Q)(D_x)F_\kappa.$$

Their squared original norm is Q^*SQ . Thus the calculation identifies the minimizer, its exact source realization, its observation, and the full relation space, not only an abstract Schur complement.

In quotient-algebra terms the second stage is $E_H/((\kappa)/(H)) = E_\kappa$. The source relation space contains all old H relations since (H) is contained in (κ) . More explicitly, the orthogonal decomposition at the source level is

$$\kappa P_{N-K} = \kappa R_h(E_h) + HP_{N-D},$$

with the two summands orthogonal for w_H by the defining minimizer property of R_h and the identity $w_h = |\kappa|^2 w_H$. This is another direct explanation of why the Schur minimization removes all and only the allowed old directions and original relation directions.

Since $\det U = \text{Res}(\kappa, h)$, the determinant identity also gives

$$\begin{aligned} \det S &= |\text{Res}(h, \kappa)|^2 \det \Gamma, \\ \det G_H &= \det G_h \det \Gamma. \end{aligned}$$

The cancellation in the final formula is justified only after retaining both the Sylvester factor in T and the full h -unit in the quotient coordinate. Neither factor was set to one.

5. Full physical-jet conversion and its source observation

Let $U_H = M_{v_H}$, $U_h = M_{v_h}$, and $U_\kappa = M_{v_\kappa}$, all multiplication by the actual full Taylor units. The coefficient and physical-jet metrics obey

$$G_H = U_H^* G_H^{\text{jet}} U_H,$$

and the analogous identities for h and κ at their indicated cutoffs. The physical packet isomorphism is

$$J = U_H T \text{diag}(U_h^{-1}, U_\kappa^{-1}).$$

Its two columns are precisely the two extension-by-zero maps in the full CRT jet decomposition: identity on the indicated old or added block and zero on the other through every required order. This follows from $v_h = (\kappa \bmod h)(v_H \bmod h)$ and

$$v_\kappa = (h \bmod \kappa)(v_H \bmod \kappa).$$

Thus the physical mixed block is $C^{\text{jet}} = U_h^* C U_\kappa^{-1}$, and its Schur complement is

$$S^{\text{jet}} = U_\kappa^* S U_\kappa^{-1}.$$

The actual physical κ observation on the same extended source is

$$\begin{aligned} F &\mapsto J_\kappa(F(D_x)F_H) \\ &= V \pi_\kappa F, \end{aligned}$$

where $V = M_{v_H \bmod \kappa}$.

Its kernel is still κP_{N-K} , and its canonical quotient metric is

$$\Gamma_{\text{free}}^{\text{jet}} = V^{-*} \Gamma V^{-1}.$$

Because $U_\kappa = VU$, the exact Schur identity becomes $S^{\text{jet}} = \Gamma_{\text{free}}^{\text{jet}}$. In particular, in physical jet coordinates it is literally the metric for fixing the added jets and leaving the old jets free. The source remains the degree- N window generated by F_H ; it has not been replaced by the smaller window generated by F_κ .

For completeness, in the same monomial coordinate conventions, $|\det J| = 1/|\operatorname{Res}(h, \kappa)|$. Indeed the restrictions of U_H have product determinant $\det U_H$ by CRT, whereas

$$\det U_h \det U_\kappa = \det U_H \operatorname{Res}(h, \kappa) \operatorname{Res}(\kappa, h).$$

Combining with $\det T$ gives the assertion. Therefore the corresponding physical determinant identity is

$$\det G_H^{\text{jet}} = |\operatorname{Res}(h, \kappa)|^2 \det G_h^{\text{jet}} \det \Gamma_{\text{free}}^{\text{jet}}.$$

These determinant statements retain full unit transformations. They do not infer that the packet splitting is orthogonal under the canonical source metric.

6. Degenerate case and scope

If $\kappa = 1$, $K = 0$ and E_κ is zero. All second blocks, residue-free observations, and their relation-quotient spaces have dimension zero; their determinants are one. The assertion reduces to the identity on the old h packet. At the smallest cutoff $N = d - 1$, interpret the empty second polynomial source as $P_{-1} = 0$. No inverse of a nonexistent positive-dimensional block is required.

Every matrix here is an actual finite-dimensional canonical source or quotient metric, with strict positivity from the original source norm. This proof makes no inference about a completed boundary range, no uniform estimate as the cutoff or packet varies, and no new zero existence claim. All repeated-root and mass data remain in the stated polynomials, full units, source sections, and integrals.

7. Full review of the root EM1–12 proof

The complete root `MAIN_METRIC_EXTENSION.md` was read after it became available, including the actual arithmetic projection, moment-determinant calculation, mixed-cost statement, $K = 0$ case, and provenance/scope. Its displayed equations EM1–12 agree with the preceding independent derivation. The following additional root calculations were checked.

In EM6, J_H is the actual full jet map on the H -torsion submodule of the original theta quotient. Restriction of those jets to the κ block, followed by the original σ_κ , is a projection onto its κ -primary submodule. Its kernel is exactly the h -primary submodule, because the packet factors are coprime with their full orders. On a source $P(D_x)F_H$ the κ observation is precisely $(v_H \bmod \kappa)[P]_\kappa$. Thus the projection has exactly the source and unit coordinates used for Γ and its physical version; no other arithmetic cohomology group has been substituted.

The root's direct proof of $\det T$ uses the monic division basis $[1, \dots, s^{d-1}, h, \dots, hs^{K-1}]$. Its coefficient change has determinant one, and T has diagonal blocks $M_{\kappa \bmod h}$ and identity in this basis. The former has determinant $\operatorname{Res}(h, \kappa)$, including each root order. This independently verifies the sign convention in EM7 without a simple-root assumption or a choice of determinant normalization.

For the root's moment determinants, define $\Delta_{f,n}$ on the n source monomials and use $\Delta_{f,0} = 1$. At an onto cutoff $n - 1$, so $n \geq \deg f$, the monic basis of low quotient powers followed by f times relation powers has change-of-basis determinant one. The relation Gram is the original weight $|f|^2 w_f = w_1$, and its dimension is $n - \deg f$. Taking the Gram Schur complement therefore proves exactly

$$\det G_{f,n-1}^{\text{coef}} = \frac{\Delta_{f,n}}{\Delta_{1,n-\deg f}}.$$

For H at N and h at $N - K$, both relation dimensions are $N - D + 1$. Their full common denominators cancel, giving the moment ratio in EM11. Alternatively, the relation kernel in EM4 has basis κ times $1, \dots, s^{N-K}$ and weight $|\kappa|^2 w_H = w_h$. Its Gram determinant is $\Delta_{h,N-K+1}$, directly proving the same formula for $\det \Gamma_N$. Both routes retain the original w_1 seed and all unnormalized masses. The minimal cutoff $N = D - 1$ uses a zero-dimensional original relation space and determinant one, not a negative-index determinant.

Finally the operator $B^{-1}C^*A^{-1}C$ in EM12 is self-adjoint and nonnegative for the existing B form because its B -pairing is given by the Hermitian nonnegative matrix $C^*A^{-1}C$. Strict positivity of the actual Schur complement S gives every eigenvalue strictly below one. Thus the product of the exact 1-minus-eigenvalue factors is the displayed determinant ratio, and $Q^*(B - S)Q = (A^{-1}CQ)^*A(A^{-1}CQ)$ is the precise original source cost of the optimizing old component. No Euclidean self-adjointness or vanishing of C is silently assumed.

Two local notation clarifications were reported before the final seal: the determinants of U_h and U_κ must be multiplied as scalars, rather than writing a matrix product of potentially different sizes; and the generic

moment-ratio formula uses the onto cutoff $n \geq \deg f$. Every substantive calculation and every actual application already satisfies these conventions.

Final independent acceptance

The revised determinant-product phrase and the explicit condition $n \geq \deg f$ were reread on 2026-09-13. Both local notation comments above are resolved; they remain recorded only as the review history.

The exact accepted root proof is `MAIN_METRIC_EXTENSION.md`, SHA-256:

`8963772b34c853ed6e3a7931830934c1928f8864a8e667d0cf63ec19c0b4dc6a`.

Final disposition: accept EM1–12 and all accompanying scope statements, with no outstanding mathematical correction. The review verifies the actual mixed source Gram, the quotient and its exact minimizer behind the Schur complement, the full resultant and physical-unit conversions, the original moment-determinant formula, the mixed source cost, and the empty-added-packet case. It certifies written mathematics only, with no numerical or Lean replay, no new zero-location or existence claim, and no uniform estimate beyond the displayed exact finite identities.